

Production capacity

The capacity of an operation is the maximum level of value-added activity over a period of time that the process can achieve under normal operating conditions (Slack et al.)

Air-conditioner plant – Number of units per week (output) • Brewery – Litres of beer produced per month (output) • Steel mill – Tonnes per hour (output) • Electricity company – Megawatts of electricity generated (output) • Hospital – Number of beds available (input) • Theatre – Number of seats (input) • University – Number of students (input) •

The capacity of a business depends on factors such as:

The physical infrastructure available (machines, workstations, space). • Workforce capabilities (number of employees, skills, shifts). • Operational efficiency (setup times, workflow management). • Market demand •

A factory produces three types of air conditioning (AC) units: Deluxe Model – Takes 1.5 hours to assemble¹. Standard Model – Takes 1 hour to assemble². Economy Model – Takes 0.5 hours to assemble³.

The factory has 800 staff hours per week available for assembly.

Case 1: Demand ratio of 2:3:2 Total time required per 7 units:

Time per unit:

Total capacity:

Case 2: Demand ratio of 1:2:4 Total time required per 7 units:

Time per unit:

Total capacity:

Product mix significantly affects capacity. Producing more of the models that require less assembly time increases total production output.

What is capacity?

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Theoretical capacity is the maximum possible output rate, whereas the actual capacity is a realistic estimate of the achievable output rate. The main difficulties which inhibit perfect utilisation of a manufacturing system, i.e., which restrict the capacity are:

MANAGE Lost Time (Planned):

Setup times • Switchover delays •

Imbalances:

Bottlenecks • Imbalances in task times •

MITIGATE Lost Time (Unplanned):

Breakdowns • Coordination conflicts (of equipment and labour)

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Supply shortages •

Reduced Yield:

Quality problems •

Variable Conditions:

Variability in process times causing a build-up of inventory

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Variability in raw material arrivals • Variability in order arrivals • Unplanned downtime •

Actual capacity is always lower than theoretical capacity due to these inefficiencies.

Bottlenecks

A bottleneck is the slowest task in the process that limits the entire system's output. "A chain is only as strong as its weakest link." If one workstation takes longer than the others, it determines the overall system's capacity.

Solution Approaches:

Increase resources at the bottleneck. • Reduce setup times or improve workflow to optimise the bottleneck's efficiency. • Balance workloads to prevent bottlenecks. •

Set-up time

The time required to prepare a machine for a different task (e.g., changing a mould in a factory). Longer setup times reduce the time available for actual production. Reducing Set-up Times:

Reduces batch size constraints (making it easier to shift between products). • Increases flexibility (more responsive to demand changes). • Improves productivity (more production in the same time frame). • Reduces waste (as defects due to changeovers decrease). •

Theoretical vs actual capacity

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Scenario: A Factory with Four Machines (A, B, C, D) The factory produces two different products:

Product 1: Uses machines $A \rightarrow B \rightarrow D$. Product 2: Uses machines $A \rightarrow C \rightarrow D$.

Whenever a batch of Product 1 is produced, an identical batch of Product 2 is made immediately afterward. Each machine has both process time (time taken per unit) and setup time (time required to prepare for production after a changeover).

Machine Process Time (Seconds per Unit) Setup Time (Seconds per Batch)

A 20 600

B 50 1000

C 70 300

D 30 400

Each batch consists of 10 or 100 units.

Case 1: Batch Size = 10 Number of products processed in one batch:

A: 20 units (since it processes both products). B: 10 units (Product 1 only). C: 10 units (Product 2 only). D: 20 units (since both products pass through it).

Total Time per Batch (Setup + Process Time)

Machine Setup Time Process Time per 10 units Total Time per Batch

A 1200s 400s 1600s

B 1000s 500s 1500s

C 300s 700s 1000s

D 800s 600s 1400s

Bottleneck (Slowest Stage): Machine B $\rightarrow$ 22.5 units per hour

Machine B is the bottleneck since it has the longest total batch time. High setup times reduce effective production capacity. Long setup times lead to larger batch sizes (since producing small batches would waste too much time on setups).

Case 2: Batch Size = 100 Number of products processed in one batch:

A: 200 units B: 100 units C: 100 units D: 200 units

Total Time per Batch (Setup + Process Time)

Machine Setup Time Process Time per 100 units Total Time per Batch

A 1200s 4000s 5200s

B 1000s 5000s 6000s

C 300s 7000s 7300s

D 800s 6000s 6800s

Bottleneck (Slowest Stage): Machine C $\rightarrow$ 49.3 units per hour

With a larger batch size, Machine C becomes the bottleneck (since it has a long process time). Larger batches reduce the relative impact of setup times. Larger batch sizes increase total production capacity (since fewer setups are needed per unit).

To improve capacity, we reduce setup times by 50% and keep batch size = 10.

Revised Setup Times (50% Reduction)

Machine New Setup Time Process Time per 10 units Total Time per Batch

A 600s 400s 1000s

B 500s 500s 1000s

C 150s 700s 850s

D 400s 600s 1000s

New Bottleneck: Balanced capacity (multiple bottlenecks) → 36 units per hour

Reduced setup times lead to more balanced operations. Capacity increases significantly from 22.5 to 36 units per hour. Smaller batches become more viable, improving flexibility.

Long setup times force larger batch sizes (to avoid excessive downtime).• Larger batch sizes limit flexibility (in responding to demand changes).• Reducing setup times increases production efficiency and flexibility. • A balanced system without excessive bottlenecks improves overall capacity. •

Breakdowns

Machine breakdowns disrupt production and cause inefficiencies.

If each workstation has a 1 in 1000 chance of breaking down at any moment: Probability of 100 machines all working simultaneously:

9.5% of production is likely to be delayed or lost.

Hence the importance of breakdown free production and routine maintenance The effect is exaggerated if a new setup is required after an interruption [a similar but more compounded analysis applies for defective parts]. Routine maintenance and preventive strategies reduce unexpected breakdowns.

Measuring capacity performance

Overall Equipment Effectiveness (OEE) measures how efficiently equipment is used, based on:

Availability Rate AR Percentage of time equipment is available.

Performance Rate PR Efficiency of output vs. potential output.

Quality Rate QR Percentage of defect-free products.

Formula:

Related capacity definitions

Capacity The capacity of an operation is the maximum level of value-added activity over a period of time that the process can achieve under normal operating conditions (Slack et al.)

Utilisation Measure of the number or % of hours worked by equipment, line, staff, etc. (Hill)

Efficiency Comparing actual output to the level of output expected (Hill)

Capacity Planning

The task of setting the effective capacity of the operation so that it can respond to demand

Challenges for capacity planning:

Capacity is a soft, malleable constraint • Capacity depends on everything • Capacity frictions: lead-times, lumpiness, fixed costs • Capacity requires large and irreversible investment •

Capacity decisions can be political • Measuring and valuing capacity shortfall is not obvious •

Capacity investment involves long-run planning under uncertainty

Arguably the greatest challenge for capacity strategy ○

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Sizing How much capacity to invest in?

Timing When to increase or reduce resources?

Type What kinds of resources are best?

Location Where should resources be located?

Ideal vs real demand

Ideal Demand:

Smooth and predictable demand is preferred in capacity planning. In an ideal scenario:

Any changes in demand are perfectly forecasted with enough time to adjust capacity.

Challenges with real demand:

In practice, demand is rarely predictable. • Real demand fluctuates due to various factors:

Seasonality: Higher demand during specific periods (e.g., summer, holidays). ○

Time-based variations: Daily, weekly, and monthly fluctuations (e.g., restaurants at lunchtime).

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Market conditions: Competitor actions, economic conditions, or product life cycle changes.

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Capacity Planning

The process of determining the production capacity needed by an organization to meet changing demands for its products.

Since demand fluctuates, businesses must carefully choose between:

Long-term capacity investments (which are costly and difficult to reverse). • Short-term flexibility measures (e.g., overtime, outsourcing) •

Capacity planning involves a three-step process: Measure aggregate demand and capacity

Use forecasting techniques to estimate demand. ○

Assess current production capacity to determine if it meets forecasted demand. ○

1.

Identify alternative capacity plans Consider different ways to adjust capacity based on cost, feasibility, and risk. ○

2.

Choose the most appropriate plan Select the best capacity plan based on business objectives. ○

3.

Capacity planning and control

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Implement and monitor the plan, making adjustments when necessary. ○

Long-term capacity planning

Long-term capacity planning considers strategic investments to ensure capacity aligns with future demand.

Three basic strategies for long-term capacity planning:

Strategy Description Advantages Disadvantages

Capacity

lead

Expanding capacity before demand increases.

Prevents shortages • Can accommodate sudden demand surges.

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Helps establish market dominance.

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Risk of overcapacity if demand doesn't grow as expected.

-

Requires early capital investment.

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Possible underutilisation.

-

Capacity

lag

Expanding capacity after demand increases.

Avoids overcapacity. • Delays capital spending, making it less risky.

- May cause customer dissatisfaction due to supply shortages.

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Higher risk of lost sales.

-

Smoothing

with inventory

Using inventory as a buffer to handle demand fluctuations.

Allows continuous production.

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Helps manage demand fluctuations without immediate capacity expansion.

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Works only for non-perishable goods.

-

Holding excess inventory is costly.

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The best strategy depends on demand predictability, cost structure, and competitive pressures. Some industries (e.g., airlines, hotels) may use hybrid strategies.

Medium-term capacity planning

Medium-term planning focuses on adjusting capacity over months or quarters. There are four primary strategies:

Level capacity plan Production remains constant regardless of demand fluctuations. Excess products are stored in inventory during low demand periods and used when demand rises.

1.

Aluminium Producer → Produces consistently throughout the year, storing excess stock.

Woollen Knitwear Factory → Produces consistently, even when demand is seasonal. Hotels

& Retail Stores → Underutilised capacity during low seasons. Advantages:

Stable workforce – Avoids frequent hiring and layoffs. High equipment utilisation –

Maximises efficiency. Lower unit costs – Producing at a constant rate reduces per-unit costs.

Disadvantages: Inventory costs – Large storage spaces are needed. Mismatch in demand and

supply – Risk of overproduction during low demand periods. Not ideal for perishable goods –

Storage is costly or impossible (e.g., fresh food).

Chase demand plan Capacity adjusts to match demand fluctuations. No inventory is carried forward; production changes as demand changes.

Aluminium Producer: Production fluctuates based on orders. Retail Stores & Hotels: Staff scheduling varies depending on peak/off-peak seasons. Advantages:

No excess inventory – Reduces holding costs. Meets customer demand directly – Avoids lost sales due to shortages.

Disadvantages: Frequent hiring and layoffs – Can be costly and affect workforce morale.

Operational inefficiencies – Rapid changes in production may reduce productivity. Higher

costs for sudden adjustments – Labour overtime, temporary workers, and equipment adjustments.

2.

Optimal capacity plan Mathematical programming is used to find the best or

3.

a decision or set of decisions about how best to use a set limited resources in order to achieve a state goal of objectives.

A Linear Programming (LP) model seeks to maximise or minimise a linear objective function, subject to a set of linear constraints, by determining values of nonnegative real decision variables.

Demand management Instead of changing capacity, companies try to shape demand through pricing and marketing strategies.

Methods: Price adjustments:

Increase price to reduce demand (e.g., luxury goods). □ Lower prices to boost demand (e.g., discounting perishable food before expiry).

□

▪

Yield Management:

Overbooking strategy (e.g., airlines selling more tickets than seats available).

□

Variable pricing based on demand (e.g., hotel room rates vary by season).

□

▪

Counter-cyclical products:

Offering alternative products to balance seasonal demand. □ Example: A ski resort offering summer hiking packages. □

▪

4.

Short-term capacity planning

Short-term capacity planning focuses on immediate adjustments within days or weeks to meet demand fluctuations without major capital investment.

Order mix management Prioritising high-margin or time-sensitive orders over standard production. Example: A custom furniture company prioritises luxury orders over bulk orders during peak periods.

1.

Scheduling downtime and maintenance efficiently Preventative maintenance helps avoid unexpected breakdowns that disrupt production. Maintenance should be scheduled during off-peak hours. Example: A car assembly plant schedules equipment maintenance on weekends when demand is lower.

2.

Overtime and extra shifts Temporary measure to increase capacity without hiring more employees. Effective when demand surges are short-term. Disadvantages:

Higher labour costs due to overtime pay. Worker fatigue can reduce productivity and quality.

3.

Hiring temporary workers or subcontracting When demand is higher than expected, companies may:

4.

Hire temporary staff (e.g., seasonal retail workers). Outsource production to third-party manufacturers. Example: A clothing brand outsources production during peak holiday seasons to meet demand.

Using inventory as a buffer Businesses producing non-perishable goods can increase stock levels before peak demand. Example: A soft drinks company builds up inventory before summer.

5.

Outsourcing and third-party manufacturing If production demand exceeds internal capacity, companies may use third-party suppliers. Example: Tech companies outsource chip manufacturing to suppliers like TSMC.

6.

Short-term capacity planning is about flexibility. Businesses must balance cost, time, and workforce well-being to manage demand fluctuations effectively.

Background of DensePak

Company Type Disk drive manufacturer.

Objective Develop a cost-effective production plan.

Challenge Fluctuating monthly demand requires either inventory storage or labour adjustments.

Key Decision Should the company use a Level, Chase, or Optimised (LP) strategy?

Key data for planning

Starting inventory 500 units

Desired ending inventory 600 units

Monthly Forecasted Demand (Units)

Month January February March April May June

Forecast Demand 1,280 640 900 1,200 2,000 1,400

of Workdays per Month 20 24 18 26 22 15

Units/Worker 2.931 3.517 2.638 3.810 3.224 2.198

Each worker produces 0.14653 disk drives per day. Cost Structure

Hiring Cost £500 per worker

Layoff Cost £1,000 per worker

Inventory Holding Cost £80 per unit per month

Current Workforce 300 workers

Option 1: level strategy

Maintains a constant workforce throughout the period. Extra production is stored as inventory for months when demand exceeds production.

Net demand over 6 months = 7,520 units. Required workforce (to produce evenly over 6 months):

Monthly demand 1,253 units

Required workers 411 (rounded up)

Production Plan Workforce remains constant at 411

Excess production is stored in inventory for future months.

Month Net Demand

Cumulative Net Demand (A)

Units / Worker

Cumulative # of Units Produced per Worker (B)

Ratio A/B (round up)

Jan 780 780 2.931 2.931 267

Feb 640 1420 3.517 6.448 221

Case study

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Mar 900 2320 2.638 9.086 256

Apr 1200 3520 3.810 12.896 273

May 2000 5520 3.224 16.120 343

June 2000 7520 2.198 18.318 411

Inventory levels for constant-workforce plan

Month # Units Produced per Worker (A)

Monthly Production (B= A*411)

Cum. Production (C) Cum.

Net Demand (D)

Ending Inventory (E=C-D)

Jan 2.931 1205 1205 780 425

Feb 3.517 1445 2650 1420 1230

Mar 2.638 1084 3734 2320 1414

Apr 3.810 1566 5300 3520 1780

May 3.224 1325 6625 5520 1105

June 2.198 903 7528 7520 8

Total 5962

Cost Breakdown

Cost Type Value

Hiring Cost (111 workers x £500) £55,500

Inventory Holding Cost (5,962 units x £80) £476,960

Total Cost £532,460

Adjusted Cost (Final Inventory 600 x £80) £580,460

Advantages: Stable workforce – Avoids frequent hiring/layoffs. High production efficiency – Machines run at constant capacity.

Disadvantages: High inventory costs – Holding costs are significant (£476,960). Risk of obsolete inventory – If demand patterns shift, excess inventory might not sell.

Option 2: chase strategy

The workforce adjusts to demand each month. No excess inventory is carried.

Month Demand Workers needed Workers hired Workers laid off

January 780 267 0 33

February 640 182 0 85

March 900 342 160 0

April 1,200 315 0 27

May 2,000 621 306 0

June 2,000 910 289 0

Cost Breakdown

Cost Type Value

Hiring Cost (755 workers x £500) £377,500

Layoff Cost (145 workers x £1,000) £145,000

Inventory Holding Cost (30 units x £80) £2,400

Total Cost £524,900

Adjusted Cost (Final Inventory 600 x £80) £572,900

Advantages: Minimal inventory costs – Only 30 units held. Production closely matches demand – Reduces excess production.

Disadvantages: High hiring and layoff costs (£522,500). Potential workforce morale issues due to constant staff fluctuations.

Option 3: optimised (LP) strategy (non-examinable)

Uses Linear Programming (LP) to minimise cost while balancing demand, workforce, and inventory.

Instead of large hiring/layoffs, smaller monthly adjustments are made. Some months inventory is stored, but in smaller quantities. The model smooths workforce fluctuations.

Cost Type Value

Hiring Cost (465 workers x £500) £232,500

Layoff Cost (27 workers x £1,000) £27,000

Inventory Holding Cost (900 units x £80) £72,000

Total Cost £331,500

Adjusted Cost (Final Inventory 600 x £80) £379,500

Advantages: Lowest total cost (£379,500) – Savings of £200,000 over the Level Strategy.

Less disruption to workforce – Avoids extreme hiring/layoff swings.

Disadvantages More complex to implement – Requires LP modelling.

Level Strategy → Stable but expensive due to high inventory costs. Chase Strategy →

Matches demand well but has high hiring/layoff costs. LP-Optimised Strategy → Best approach with the lowest total cost.

This case study highlights how mathematical modelling (LP) can optimise capacity planning, balancing costs, workforce stability, and production efficiency.

Conflicting objectives:

Supply

100% utilisation of resources○

Zero stock○

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Demand

Any volume and product mix○

No lead time○

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This requires coordination within and between firms aiming at:

Balanced flow of work - all tasks take the same time • Steady demand for products - and hence for inputs •

Ideal demand is smooth and predictable:

Total demand = maximum output capacity of resources • Any changes are perfectly forecast in sufficient time to allow capacity change •

But ...

Real demand is usually not predictable • Demand has peaks -

lunchtime/Saturday/summer/etc. • Demand varies through product life cycle and competition •

Order fulfilment process

From “order to delivery” (OTD) • Key process, day-to-day challenge! • Determines customer service and majority of cost •

Order fulfilment strategy

How far back does the order go into the supply chain? • Which is appropriate under which circumstances? •

Two basic factors are important:

Production Lead-time P: How long does it take to make the product? • Demand Lead-time D:

How long is the customer willing to wait for the product? •

The P:D ratio

What is the P:D Ratio?

P (Production Lead Time): The time required to manufacture a product from start to finish.

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D (Demand Lead Time): The time customers are willing to wait for the product. •

If $P < D$, production can begin after an order is placed.

Order fulfilment strategies

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P:D Ratio Strategy Explanation Outcome Example

$P \ll D$ (Production is much faster than demand lead time)

Make-to-Stock (MTS)

Goods made to be placed in stock prior to receiving an order; typical of commodities and continuous processing, some line production

Efficient production

Risk of obsolescence / high stock cost

Supermarkets stock everyday items in advance.

$P > D$ (Production takes longer than customers are willing to wait)

Assemble-to-Order (ATO)

Producers hold components stock to assemble an order as required by the customer; typical of line/ batch production

Responsive for customised products

Cost of stock

Dell assembles computers after an order but pre-stocks components.

$P \gg D$ (Production is significantly longer than demand lead time)

Make-to-Order (MTO)

Material ordered and product or service made only after the buyers order is received. Line, batch and job shop production

No FGI cost

Less efficient production

Boeing only starts aircraft production after receiving customer orders.

$P \gg D$ (Production involves custom engineering)

Engineer-to-Order (ETO)

Product designed and built to customer order; typical of projects, some job shop

Specialised machinery or construction projects designed from scratch.

A decoupling point is where inventory is strategically stored to balance production efficiency with customer responsiveness.

Upstream Decoupling → Closer to raw materials (e.g., steel for car manufacturing).•

Downstream Decoupling → Closer to finished goods (e.g., pre-built cars in dealerships).

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Decoupling Point Fulfilment Strategy Example

Finished Goods Inventory (FGI)

Make-to-Stock (MTS)

FMCG products (e.g., Coca-Cola, toothpaste)

Work-in-Process (WIP) Assemble-to-Order (ATO)

Dell pre-stocking parts for rapid assembly

Raw Materials & Components

Make-to-Order (MTO)

Aircraft manufacturers waiting for confirmed orders

Custom Engineering Design

Engineer-to-Order (ETO)

Bridge construction tailored to site specifications

Dell's Build-to-Order (BTO) Model

P:D Ratio → P is longer than D, so Dell cannot wait until orders arrive to begin production.

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Decoupling Point → Component level, allowing rapid assembly when orders are placed.

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The P:D ratio determines the best fulfilment strategy (MTS, ATO, MTO, or ETO).

Decoupling points help balance efficiency and responsiveness by storing inventory at the right stage in the supply chain. Companies like Dell (ATO) and Zara (Real-Time Demand) optimise decoupling to stay competitive.

Mass customisation

Mass Customisation is the ability to produce customised products while maintaining the efficiency and cost advantages of mass production. It combines the flexibility of custom manufacturing with the high-volume production of standardised goods.

Customer Involvement – Buyers choose features, materials, or designs. •

Modular Components – Products are built using interchangeable parts. •

Flexible Production Systems – Automated manufacturing adapts to different configurations.

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Efficient Supply Chain – Customisation occurs late in the production process (delayed differentiation).

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Type Description Example

Collaborative Customers co-design their product Nike ID (Custom trainers)

Adaptive Standard product with features that users adjust

Adjustable car seats

Cosmetic Same product, different branding for different markets

Coca-Cola bottles with different names

Transparent Customisation happens without customers realising

Amazon's personalised recommendations

Examples:

Dell Computers – Customers configure laptops before purchase (Assemble-to-Order).•

Nike By You (Nike ID) – Personalised trainers with chosen colours and initials. •

Levi's Custom Jeans – Tailored fit options available online. •

BMW – Custom vehicle configurations built on a modular platform. •

Customisation without losing efficiency → Achieved through modular designs.

Delayed differentiation → Customisation happens at the last possible stage.

Technology-driven → AI, robotics, and supply chain integration make mass customisation feasible.

Mass customisation is closely linked to the P:D ratio because it allows companies to offer personalised products without long delays.

BMW's Custom Car Manufacturing

P (Production Lead Time) $\gg$ D (Customer Wait Time) → Customers want custom cars but won't wait months.

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Solution: Assemble-to-Order (ATO) with a modular platform.

BMW pre-manufactures standard parts (engines, chassis, electronics).○

Customers choose features (paint, interior, wheels).○

Custom assembly is completed quickly → Decoupling Point: WIP Inventory (Partially Built Cars).

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Mass Customisation works best when customisation happens late in the process (postponement strategy).

Forecasting is the process of predicting future demand based on historical data, trends, and external factors. Used in operations, supply chain management, finance, and marketing.

Helps companies make data-driven decisions to optimise inventory, production, and staffing.

Forecasting is essential for:

Operations planning → Ensures the right resources are available at the right time. • Capacity

management → Helps prevent overproduction or underproduction. • Inventory control →

Reduces excess stock and stockouts. • Financial planning → Supports budgeting and cost control. •

A good forecast reduces uncertainty and allows businesses to be proactive rather than reactive.

Demand patterns are influenced by:

Seasonality → Sales vary based on time of year (e.g., ice cream in summer). • Trends →

Long-term shifts in consumer behaviour (e.g., electric cars). • Cyclical variations →

Repeating up-and-down trends over multiple years (e.g., economic cycles).

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Random noise → Short-term fluctuations caused by unpredictable events (e.g., weather changes).

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A retailer sees higher demand for winter coats every November–January due to seasonality.

Independent and dependent demand

Understanding the difference between independent and dependent demand is crucial for effective inventory management and forecasting.

Independent Demand

Demand that is determined by external market conditions rather than internal

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Forecasting

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processes.

Characteristics:

Unpredictable and fluctuating (requires forecasting). ▪

Not directly linked to other demand within the business. ▪

Must be managed with safety stock and demand forecasting. ▪

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Examples:

Finished goods (e.g., cars, mobile phones, laptops). ▪

Retail products (e.g., groceries, clothing, furniture). ▪

Service demands (e.g., airline tickets, hotel bookings). ▪

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Forecasting Method Used:

Requires forecasting models (e.g., moving averages, exponential smoothing).

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Example Scenario:

A car dealership needs to predict the number of cars customers will purchase next month.

This demand is independent because it is driven by market trends, customer preferences, and economic conditions.

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Dependent demand

Demand that is derived from the demand for another item (typically a finished product).

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Characteristics:

Predictable and structured (does not require traditional forecasting).▪

Directly linked to independent demand.▪

Managed using Materials Requirement Planning (MRP) systems.▪

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Examples:

Car components (e.g., tyres, engines, brake pads).▪

Computer parts (e.g., processors, RAM, keyboards).▪

Raw materials (e.g., steel for construction, flour for baking).▪

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Planning Method Used:

Managed using Bills of Materials (BOM) and MRP software.▪

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Example Scenario:

A car manufacturer forecasts demand for cars (independent demand) and then calculates how many engines, tyres, and seats are needed (dependent demand).

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Factor Independent Dependent

Definition Market-driven demand for finished goods.

Demand for components based on production needs.

Predictability Unpredictable, requires forecasting.

Predictable, based on production schedules.

Example Cars, retail goods, airline tickets. Car tyres, laptop batteries, raw materials.

Management Approach

Forecasting techniques (e.g., moving averages).

Material Requirements Planning (MRP).

Independent demand requires forecasting, while dependent demand is calculated based on production needs.

Qualitative vs. quantitative forecasting

Forecasting methods are generally divided into qualitative and quantitative techniques.

Qualitative forecasting:

A subjective approach that relies on expert opinions, intuition, and market research rather than numerical data.

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Best Used For:

New product launches (no historical data).○

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Long-term strategic planning (5+ years ahead).○

Industries with rapidly changing trends (fashion, technology).○

Common Qualitative Methods:

Method Description Example

Delphi Method

Experts provide anonymous forecasts, refining predictions through multiple rounds.

Predicting demand for new energy sources (solar, hydrogen).

Market Surveys

Collecting customer feedback via surveys and focus groups.

Determining customer interest in a new smartphone model.

Sales Force Opinions

Sales teams estimate future demand based on customer interactions.

Forecasting demand for new car models based on dealership inquiries.

Executive Judgement

Senior managers make forecasts based on experience and intuition.

A CEO predicting the next industry trend without numerical data.

Advantages of Qualitative Forecasting:

Useful when historical data is unavailable. • Incorporates expert judgement and market insights. • Can predict disruptive innovations. •

Disadvantages:

Subjective and biased (opinions can be inaccurate). • Not useful for short-term operational decisions. • Difficult to test for accuracy. •

Best for new products, strategic decisions, and long-term trends, but not precise for short-term planning.

Quantitative forecasting:

A data-driven approach that uses historical sales data and statistical models to predict future demand.

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Best Used For:

Established products with historical sales data.○

Short- to medium-term planning (1–24 months).○

Stable industries with measurable trends (retail, manufacturing).○

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Common Quantitative Methods:

Method Description Example Usage

Moving Averages

Calculates the average demand over a set number of past periods to smooth fluctuations.

Forecasting weekly bread sales at a bakery.

Exponential Smoothing

Assigns more weight to recent data, making forecasts more responsive.

Predicting next month's demand for mobile phones.

Regression Analysis

Examines relationships between dependent and independent variables.

Forecasting car sales based on fuel prices.

Time Series Analysis

Identifies patterns in historical data, including seasonality and trends.

Predicting Christmas toy sales.

Advantages of Quantitative Forecasting:

Objective and based on actual data. • Accurate for short- to medium-term forecasts. • Easier to validate and test for errors. •

Disadvantages:

Assumes past trends continue (not always true). • Requires clean and accurate data. • Struggles with sudden market changes (e.g., COVID-19 disruptions). •

Best for short- to medium-term demand planning in stable industries, but less effective for predicting major market shifts.

Comparison of Qualitative vs. Quantitative Forecasting:

Factor	Qualitative	Quantitative
Data required	No historical data needed	Requires historical data
Forecasting horizon	Long-term (1–5 years+)	Short- to medium-term (1–24 months)
Precision	Subjective, less precise	Data-driven, more precise
Best Used For	New products, market disruptions, strategic planning	Stable demand patterns, inventory & supply chain planning
Example	Predicting future AI adoption trends	Forecasting next month's laptop sales

Use qualitative methods for long-term predictions and new product launches. • Use quantitative methods for short-term operational planning and inventory management.

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Time series vs. causal models

Time Series Forecasting

Uses historical data to predict the future. • Assumes that past trends continue. • Common methods:

Moving Averages (Simple & Weighted). ○

Exponential Smoothing (Single, Double, Triple). ○

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Example: A coffee shop predicts next month's sales based on the last 12 months. • Causal models

Forecasts demand based on relationships between variables. • Uses external factors like economic indicators, weather, or marketing spend. • Common methods:

Regression Analysis○

Machine Learning Models○

•

Example: Ice cream sales depend on temperature – hotter days lead to higher sales. •

Moving Average Method

A Moving Average (MA) is a simple forecasting method that calculates the average demand over a specific number of past periods to predict future demand. It is useful for smoothing out random fluctuations and identifying underlying trends in data.

Types of Moving Averages:

Simple Moving Average (SMA) – Gives equal weight to all past observations. • Weighted

Moving Average (WMA) – Assigns more weight to recent observations. •

Formula for Simple Moving Average (SMA)

Where:

S_t Forecast for the next period

x_t Actual demand in period t

n Number of periods in the moving average

By convention, S_t is an average based on data up to time t , but used as a forecast for time $t + 1$.

Example Calculation (4-Period SMA):

Month Sales (Units)

Jan 100

Feb 120

Mar 110

Apr 130

May (Forecast) ?

Using a 4-month moving average:

Thus, the forecasted demand for May is 115 units.

The more data periods are used, the older the data gets:

1 period MA: average age of data is:

1 period

•

2 period MA: average age is:

•

3 period MA: average age is:

•

n period MA: average age is:

•

In a MA forecast, each period has a weight of $1/n$; i.e., the oldest data has the same weight as the most recent.

In WMA, more recent data is given higher importance. Weighted Moving Average with 3 Periods:

Month Sales (Units) Weight

Mar 110 0.2

Apr 130 0.3

May 150 0.5

Thus, the forecast for June is 136 units.

Advantages:

Simple to calculate and understand. • Effective for stable demand patterns. • Smooths out random fluctuations. •

Disadvantages:

Equal weighting (SMA) can be unrealistic – recent data may be more relevant. • Lagging effect – does not respond quickly to sudden demand shifts. • Not suitable for trend or seasonal demand. •

Best for: Stable demand with minor fluctuations.

Exponential smoothing method

Exponential Smoothing (ES) is an advanced method of forecasting that assigns more weight to recent observations while still considering past data. Unlike Moving Averages, which give equal weight to past values, Exponential Smoothing gradually decreases weights for older data.

Formula for Single Exponential Smoothing (SES)

Where:

S_t Smoothed forecast for the next period

x_t Actual demand in period t

S_t

S_{t-1} Forecast from the previous period

α Smoothing constant ($0 < \alpha < 1$)

Like a moving average, but weight given to each period is a fixed proportion of weight given to succeeding period:

Choosing the Right α Value

$0 < \alpha < 1$, so weights get smaller as we recede into the past

Higher α (e.g., 0.8): Reacts quickly to recent demand changes. Use higher α for volatile demand.

•

Lower α (e.g., 0.2): Produces a stable forecast but slower response. Use lower α for steady demand.

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The constant k is chosen so that the weights sum to 1:

•

And this (it can be shown) requires $k = \alpha$

It can be shown that the average age of data = $1/\alpha$

Forecast for period $(t+1)$ can be quickly calculated from the forecast for period t :

Alternative notations:

Where:

Example Calculation (SES with $\alpha = 0.3$)

Month Actual Sales (x_t) Forecast (S_t)

Jan 100 100 (Initial)

Feb 120 ?

Mar 110 ?

Apr 130 ?

Forecast for February

1.

Forecast for April

3.

For a step change in demand:

For “Ramp” (linear trend) demand:

For a “Parabolic” input:

Double exponential smoothing (Holt’s Method)

Basic idea is to introduce a trend estimate. Similar to single exponential smoothing but we have two equations. Need to choose two smoothing rates, α and β . Trend dominates after a few periods in forecasts so forecasts are only good for a short term.

Trend adjusted level equation

Trend equation

Forecast

Used when demand shows an upward or downward trend (e.g., sales growth).

Triple exponential smoothing (Winter’s Method)

Basic idea is to introduce a trend estimate and a seasonality estimate. Similar to double exponential smoothing but we have three equations. Need to choose three smoothing rates, α , β , and γ . Forecasts are only good for a short term.

Level adjustment

Trend adjustment

Seasonality adjustment

Used for forecasting sales with seasonal cycles (e.g., retail, tourism).

Strengths & weaknesses of exponential smoothing Advantages:

More accurate than Moving Averages because it reacts to recent data. • Adjustable sensitivity through α . • Can be expanded to handle trends (Holt’s) and seasonality (Winter’s). •

Disadvantages:

Choosing the right α is tricky (trial and error). • Single Exponential Smoothing struggles with trends. • Cannot predict sudden external shocks (e.g., economic crises). •

Best for: Short- to medium-term forecasts, especially when demand fluctuates.

Summary of extrapolation forecast models:

Factor Moving Average (MA) Exponential Smoothing (ES)

Calculation Simple average of past data Weighted sum with recent data given more weight

Responsiveness Slow to react to changes Faster reaction to changes

Best Used For Stable demand with no trend/seasonality

Fluctuating demand with short-term changes

Handles Trends?

No Yes (with Holt’s)

Handles Seasonality?

No Yes (with Winter's)

Complexity Easy Moderate (single ES), Complex (triple ES)

MA and single ES work well for models with constant mean demand (or irregular step changes)

-

Double ES works well for models with linear trends • Triple ES works well for models with quadratic trends (but these are rare in practice) •

Further approaches to forecasting:

One approach is to extrapolate past data (MA, ES)

This works fine for minor variability ○

Problem: response to step changes not immediate, forecast may be late in reacting to trends

-

-

Another approach is to learn about the underlying properties of the demand time series

-

How to analyse the underlying demand pattern:

Regression analysis: what factors matter? E.g., weather and sporting events to predict beer consumption

-

Decomposition/Fourier analysis: the idea is that one complex demand time series can be decomposed into a set of simpler series

-

-

Decomposition Models:

Additive Model:

-

Multiplicative Model:

-

Cyclic component is like the seasonal component, but with a longer cycle period: simple models ignore it

-

Fourier analysis:

Forecasting errors & accuracy

Forecasting is never 100% accurate, so businesses need to measure and manage errors to improve predictions. Understanding forecasting accuracy helps organisations adjust models, reduce uncertainty, and optimise inventory and operations.

Why measure forecasting accuracy?

A good forecast should:

Minimise deviations from actual demand.○

Improve decision-making (e.g., production planning, stock levels).○

Reduce costs (e.g., lower stockouts and excess inventory).○

•

A poor forecast leads to:

Stockouts → Lost sales & unhappy customers.○

Excess inventory → High holding costs & obsolescence.○

Inefficient labour scheduling → Overstaffing or understaffing.○

•

Forecast error is the difference between actual demand and forecasted demand:

There are several ways to measure forecasting accuracy:

Mean Absolute Deviation (MAD)

Measures the average absolute error between actual and forecasted values.○

Formula:

•

•

Where:

A_t Actual demand at time t

F_t Forecasted demand at time t

n Number of periods

Example calculation:

Month Actual Demand (A_t) Forecast (F_t) Absolute Error ($|A_t - F_t|$)

Jan 120 115 5

Feb 140 135 5

Mar 110 120 10

Apr 130 125 5

Total 25

•

Interpretation:

Lower MAD = More accurate forecast.▪

Higher MAD = Larger forecasting errors.▪

•

Best for: Understanding absolute error size without directional bias.○

Mean Squared Error (MSE)

Measures the average squared error, penalising large errors more than small ones.○

Formula:

•

Example calculation:

Month Actual Demand (A_t)

Forecast (F_t)

Error ($A_t - F_t$)

Squared Error ($(A_t - F_t)^2$)

Jan 120 115 5 25

Feb 140 135 5 25

Mar 110 120 10 100

Apr 130 125 5 25

Total 175

•

Interpretation:

Large errors are penalised more heavily than in MAD.▪

Difficult to interpret since the units are squared.▪

•

Best for: Situations where avoiding large errors is critical (e.g., financial forecasting).

•

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Mean Absolute Percentage Error (MAPE)

Expresses forecasting error as a percentage of actual demand, making it easier to compare across different scales.

•

Formula:

•

Example calculation:

Month Actual demand (A_t)

Forecast (F_t)

Absolute percentage error ($(A_t - F_t)/A_t \times 100$)

•

-

Jan 120 115 4.17%
Feb 140 135 3.57%
Mar 110 120 9.09%
Apr 130 125 3.85%
Total 20.68%

Interpretation:

Useful for comparing forecasts across different industries/products.▪

Fails when actual values are near zero (division by small numbers leads to large errors).

-

-

Best for: Retail, finance, and demand forecasting where percentage errors matter.○

Tracking Signal (TS)

Detects whether the forecast is biased (consistently over- or under-predicting demand).

-

Formula:

-

Interpretation:

If TS is within ± 4 , the forecast is unbiased.▪

If TS > 4 (positive), forecasts are underestimating demand.▪

If TS < -4 (negative), forecasts are overestimating demand.▪

-

Best for: Identifying persistent forecast bias over time.○

-

Different forecasting errors are suited for different applications.

Error metric

Best for When to avoid

MAD Measuring absolute forecasting error size When large errors need extra penalties

MSE Penalising large errors (financial forecasting)

When interpretability is important

MAPE Comparing errors across different scales (retail)

When actual demand is near zero

TS Identifying consistent forecast bias When random fluctuations dominate

Use multiple error metrics to get a well-rounded view of forecasting accuracy.

If forecasting errors are high, consider the following improvements: Improve Data Quality

Use real-time sales data.○

Remove data entry errors and missing values.○

Segment data into seasonal vs. non-seasonal demand.○

1. Choose the Right Forecasting Method Use Exponential Smoothing for short-term trends.○
Use Time Series models for seasonal fluctuations.○
Apply Regression Analysis if external factors affect demand.○
2. Adjust Forecasts Based on Error Analysis If bias is detected (Tracking Signal $> \pm 4$), adjust forecasts upward or downward.○
If demand is volatile, consider a higher smoothing constant in Exponential Smoothing.

•

If MAPE is high, review external demand factors like promotions or weather.○

3. Consequences of poor forecasting:
Too much inventory → high storage costs• Too little inventory → stockouts & lost sales•
Inefficient workforce planning and incorrect financial projections. A clothing retailer orders too many winter coats in March, leading to excess stock and discounting losses.
Alternative approach – demand-driven supply chains
Some companies avoid forecasting errors by using real-time demand tracking instead:
Zara reacts to sales data instantly, producing new clothing in just 2 weeks.• More accurate than long-term forecasts.•
Forecasting is crucial for inventory, production, and workforce planning.• No forecast is 100% accurate, but good methods reduce uncertainty.• Choosing the right method depends on demand patterns and business needs.•
Operations Management (OM) is concerned with the planning, scheduling, and control of manufacturing and service entities to ensure efficiency and effectiveness. It plays a vital role in managing processes that transform inputs into outputs.
Efficiency How well resources are utilised
Effectiveness How well objectives are achieved
Optimal OM strikes a balance between both—maximising output while meeting customer expectations.
Key Aspects of OM
Managing processes: Ensuring that operations are smooth and efficient.• Optimising resource usage: Achieving maximum productivity with minimal waste.• Meeting customer demand: Ensuring that goods and services are delivered in a timely and satisfactory manner.

•

Continuous improvement: Using process thinking to enhance efficiency and quality.•

Core Functions of OM

Planning: Designing processes that align with business goals.• Scheduling: Allocating resources effectively to optimise efficiency.• Control & Monitoring: Ensuring processes meet performance objectives.• Improvement: Implementing strategies such as Lean and Six Sigma to enhance operations.

•

OM is concerned with managing the process that converts: inputs (i.e., material, manpower, capital, information, and energy) into outputs (i.e., goods and services).

OM uses process thinking to meet and exceed customer demand while using all resources efficiently in order to maximise the value of the organisation. Anything which repeats in the operation can be seen as a process. The repetitive nature of processes allows for improvement. Distinction between operations which manufacture goods and those which provide services.

Operations can be broadly classified into:

Manufacturing operations

Involves tangible goods, standardised production processes, and inventory storage.

Service operations Involves intangible products, high customer interaction, and simultaneous production and consumption.

Despite these differences, many industries combine both manufacturing and service elements.

For example, a car manufacturer (manufacturing) provides after-sales services (service).

Goods (Tangible Products) Services (Intangible Activities)

Tangibility Physical, can be touched and stored.

Intangible, cannot be physically handled.

Storage Can be produced in advance and stored.

Consumed as they are produced; cannot be stored.

Production vs. Consumption

Produced before consumption.

Production and consumption occur simultaneously.

What is operations management

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Customer Involvement

Low involvement; standardised production.

High involvement; experience varies per customer.

Quality Measurement

Measurable (e.g., defect rates, specifications).

Subjective (depends on customer perception).

Transportability Can be shipped or transported.

Delivered at the point of service (e.g., healthcare, education).

Example Cars, clothing, smartphones. Banking, healthcare, consulting.

OM vs Supply Chain Management (SCM)

OM focuses on internal processes within a single firm. • SCM manages interactions between multiple firms across the supply chain. • Both OM and SCM are interconnected and crucial for business success. •

Why does OM matter?

Operations Management is one of the most critical business functions because:

It accounts for 70-80% of a company's employees and capital assets. • It directly impacts financial performance through cost reduction and efficiency. • It influences customer satisfaction and brand reputation. •

Operational decisions affect social, ethical, and environmental factors (e.g., sustainability, fair labour practices). Companies that excel in OM gain a competitive advantage by delivering high-quality products/services efficiently and cost-effectively.

Key operations decisions

Structural – long term decisions related to the delivery process, and the flow of goods and services

Location • Capacity • Technology • Vertical Integration •

Infrastructural – short term decisions related to planning and control systems of operations

Workforce • Quality Management • Organization Structure • Policies/Procedures •

A process is the sequence of operations and involved events, taking up time, space, expertise or other resources, which lead/(should lead) to the production of some outcome.

OM follows the Input-Transformation-Output Model, where:

Inputs (materials, labour, capital, energy, information)• Transformation process (production, service delivery, value addition)• Outputs (finished goods, services)•

Key components of the management system

Process Design: Structuring operations for efficiency.• Resource Planning: Allocating capital, materials, and labour.• Task Scheduling: Ensuring optimal resource use.• Performance

Measurement & Improvement: Monitoring and refining processes.•

A simplified manufacturing system

(FGI = finished goods inventory)

Suppliers provide raw materials and components.• Manufacturing process converts raw materials into finished products.• Customers receive goods and services.• Management System oversees planning, scheduling, and inventory.•

A well-managed transformation process ensures high efficiency and customer satisfaction.

Operations as a transformation process

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Internal objectives: “shareholder value”

100% effective use of resources• Minimal operating expenses: zero defects, zero stock•

External objectives:

Quality: doing things right, to a standard• Speed: time to satisfy order• Dependability: reliable delivery when promised• Flexibility: ability to change what is delivered• Cost: price competitiveness•

“The right product, at the right time, at the right quality, at the right price”

Performance Objective

Definition Example

Quality Doing things correctly and meeting specifications.

Zero defects in product manufacturing.

Speed Reducing time to deliver goods/services.

Fast-food chains serving meals in under 5 minutes.

Dependability Delivering as promised without delays.

Amazon’s next-day delivery service.

Flexibility Adapting to changes in demand or customisation.

Car manufacturers offering customisable vehicle features.

Cost Minimising production and operational expenses.

Reducing waste through Lean Manufacturing.

Companies often balance these objectives based on their business strategy and customer expectations. Operations decisions have significant cost implications, affecting:

Inventory Costs Cost of capital & warehousing: Storing excess inventory incurs costs.○

Handling & obsolescence: Damaged or outdated stock leads to financial losses.○

1.

Production Costs Batch size vs. production costs: Larger batches reduce cost per unit but increase inventory holding costs.

•

Machine & labour expenses: Highly automated systems require capital investment but lower long-term costs.

•

2.

Logistics & Distribution Costs Transport costs: Impacted by shipment frequency, fuel prices, and route efficiency.

-

3.

Sales & Marketing Costs Lost sales opportunity cost: Inefficiencies in operations may result in missed revenue.

-

4.

By optimising operations, companies can reduce costs and improve profitability.

The “strategic objective” of Operations Management is often only seen as to increase the efficiency in production, i.e., reduce unit cost.

Performance objectives

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The customer value equation:

Performance is a function of:

Quality: doing things right, to a standard• Speed/Dependability: reliability and speed of delivery• Flexibility: ability to change (volume, product mix, design)•

Firms differentiated by their trade-off across these performance objectives.

Competing on cost

Competing on cost (low-cost leadership)

Offering product at a low price relative to competition• Typically high volume products•

Often limit product range & offer little customization• May invest in automation to reduce unit costs• Can use lower skill labour• Low cost does not necessarily mean low quality•

A company aims to offer the lowest price while maintaining profitability. Focus on high efficiency, process optimisation, and cost reduction. Uses economies of scale, automation, and low-cost labour. Often features limited product variety and minimal customisation.

McDonald's

Uses standardised processes and automation to keep costs low while delivering fast, affordable meals.

-

IKEA

Offers flat-pack furniture, self-service, and bulk manufacturing to reduce costs and offer low prices.

-

Walmart

Uses supply chain efficiency and bulk purchasing to maintain low retail prices.

-

Competing on quality

Superior products & services - emphasis on high-quality materials, precise manufacturing, and excellent customer service.

Quality is often subjective• Quality is defined differently depending on who is defining it•

Service quality is meeting and exceeding customer requirements• High quality services companies demonstrate:

Superior product features○

Excellent customer service○

Consistent delivery○

Process quality – error free delivery○

-

Can be product-based (high-end, durable goods) or service-based (superior customer experience). Higher quality may justify premium pricing.

Apple

High-quality design and premium materials in iPhones and MacBooks.

-

Toyota

Known for reliability and durability, driven by its Lean Manufacturing System.

-

Competitive priorities

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Ritz-Carlton Hotels

Exceptional customer service and luxury experiences make it a market leader in hospitality.

-

Competing on speed

Fast & efficient delivery - companies prioritise rapid order fulfilment and short lead times.

Rapid delivery

Focused on shorter time between order placement and delivery, minimal wait times

-

-

On-time delivery

Deliver product exactly when needed every time○

-

Availability

Convenient and readily available when customer requires○

-

Uses automation, technology, and streamlined supply chains to reduce delays. Speed is a major advantage in industries where time is a competitive factor (e.g., e-commerce, food delivery).

Amazon Prime

Offers same-day or next-day delivery, made possible by advanced logistics and warehouse automation.

-

FedEx & DHL

Compete by ensuring fast and reliable package delivery worldwide.

-

Domino's Pizza

Built its reputation on the promise of "30-minute delivery or your pizza is free".

-

Competing on dependability

Reliability & consistency - a company focuses on delivering its products/services on time and as promised. Ensures on-time delivery, minimal defects, and consistent performance.

Dependability builds customer trust and brand loyalty.

Toyota

Maintains consistent quality and reliability across all its vehicles.

-

Singapore Airlines

Renowned for punctuality and high service standards.

-

Amazon

Guarantees reliable delivery times, making it a trusted e-commerce leader.

-

Competing on flexibility

Adaptability to market changes - companies that compete on flexibility can quickly adjust to market demand, product customisation, and industry changes.

Company environment changes rapidly

Company must accommodate change by being flexible○

-

Easily customize product/service to meet specific requirements of a customer

Ability to ramp capacity up and down to match market demands○

-

Flexibility applies to:

Product/Service Flexibility – Ability to introduce new products quickly. • Mix Flexibility – Ability to offer a variety of products. • Volume Flexibility – Ability to scale production up or down based on demand. • Delivery Flexibility – Ability to adjust order fulfilment times to meet customer needs. •

Examples:

Nike

Allows customers to customise their shoes through its Nike By You service.

-

Zara

-

Rapidly adjusts clothing designs based on real-time customer trends and brings new styles to market in just a few weeks.

Trade-offs

Cost vs. Quality Higher quality often means higher costs

Budget airlines vs. luxury airlines

Speed vs. Cost Faster delivery requires higher logistics costs Amazon Prime vs. standard shipping

Flexibility vs. Efficiency Highly flexible operations may reduce efficiency Mass production vs. custom-built products

Some companies master multiple competitive priorities, but most must strategically choose trade-offs that align with their business goals and market expectations. World-class companies (e.g., Toyota, Amazon, Apple) have successfully integrated multiple strategies to dominate their industries.

Order Winners vs. Order Qualifiers

Order Qualifiers

Minimum criteria for customers to consider purchasing (e.g., decent product quality)

Order Winners Factors that differentiate a company from competitors (e.g., Apple's innovation)

Understanding these distinctions helps businesses develop effective operational strategies.

Performance frontiers

A performance frontier represents the maximum level of performance a company can achieve across different operational objectives given current resources, technology, and constraints. It shows the trade-offs between competing priorities, such as cost, quality, speed, dependability, and flexibility.

Companies operate within a boundary of capabilities—improving one performance aspect (e.g., quality) may reduce another (e.g., cost efficiency). Shifting the frontier requires innovation, new technologies, or process improvements to achieve higher levels of performance without trade-offs.

Types of Performance Frontiers Static (Trade-off) Performance Frontier

Represents the current best possible performance given existing resources and technology.

-

Companies must choose trade-offs—e.g., increasing flexibility might reduce efficiency.

-

Example: A low-cost airline (Ryanair) keeps fares cheap by limiting service quality and flexibility.

-

1. Dynamic (Shifting) Performance Frontier Represents an expanded capability through innovation and process improvements.○

2. Companies push the boundary, achieving higher performance without major trade-offs.

-

Example: Toyota's Lean Manufacturing enabled both high quality and low costs, shifting its performance frontier beyond competitors.

-

Illustration of Performance Frontiers:

Imagine a company can currently produce high-quality products but at high costs. This defines its static performance frontier. If it adopts automation, AI, or process improvements, it may be able to:

Maintain high quality • Reduce costs • Increase speed •

This shifts the company's performance frontier, allowing it to compete more effectively without sacrificing other priorities.

Amazon (Shifting Performance Frontier)

Traditionally, fast delivery meant higher costs. • Amazon invested in automation, AI-driven logistics, and fulfilment centres.

Result: It lowered costs while improving speed and dependability, pushing its performance frontier forward.

-

-

Tesla (Shifting Performance Frontier)

Traditionally, high-tech electric vehicles were expensive and low in volume. • Tesla invested in gigafactories, battery innovations, and software integration.

Result: It lowered EV costs while improving product flexibility and quality.○

-

Key Operations Decisions

Structural – long term decisions related to the delivery process, and the flow of goods and services:

Location• Capacity• Technology• Vertical Integration•
 Infrastructural – short term decisions related to planning and control systems of operations:
 Workforce• Quality Management• Organization Structure• Policies/Procedures•
 Pure Goods Pure Services
 Tangible Intangible
 Can be stored Cannot be stored
 Production precedes consumption Production and consumption are simultaneous
 Low customer contact High customer contact --> inherent variability
 Can be transported Cannot be transported
 Quality is evident Quality difficult to judge
 Definitions:
 “An accumulation of a commodity that will be used to satisfy future demand.”
 Johnson and Montgomery (OR profs)

•

“The stocks or items used to support production (raw materials and work-in-process items), supporting activities (maintenance, repair, and operating supplies), and customer service (finished goods and spare parts).”
 APICS (Association for Operations Management) Dictionary

•

“Dead material.”
 Taiichi Ohno (Father of the Toyota Production System)

•

“A substitute for information.”
 Michael Hammer (Process Reengineering Guru)

•

Inventory is essential for managing demand, but excess inventory can lead to inefficiencies.
 Types of inventory
 Inventory is classified based on time (where it exists in the production process) and based on function (why it is being held). Classification by Time (Stages in the Production Process)
 Raw Materials (RM) Definition: Unprocessed inputs that will be used to create a product.○
 Example: A car manufacturer’s raw materials include steel, rubber, glass, and plastic.

•

Characteristics:
 Purchased from external suppliers.▪
 The quality of raw materials affects the final product’s quality.▪
 Inventory management challenge:
 Need to balance bulk buying discounts with storage costs.□ Risk of supply chain disruptions if raw materials are delayed.□

▪

-

1.

Work-in-Process (WIP) Definition: Inventory that is in production but not yet completed.○

Example:

An automobile chassis being assembled before the engine is installed.▪

A cake in a bakery that has been baked but not yet decorated.▪

-

Characteristics:

Held between processing stages.▪

High WIP levels increase lead times and hide inefficiencies.▪

Just-in-Time (JIT) strategies aim to reduce WIP and improve flow efficiency.

-

-

2.

Finished Goods Inventory (FGI)

3.

Inventory

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Definition: Products that have been fully assembled and are ready for sale or distribution.

-

Example:

A fully built Tesla car at a dealership.▪

A packaged smartphone ready to be shipped.▪

-

Characteristics:

Directly affects customer satisfaction.▪

Companies must balance demand forecasting to avoid stockouts or excess inventory.

-

Inventory management challenge:

If demand falls, FGI may become obsolete.□ If demand rises unexpectedly, stockouts can lead to lost sales.□

-

-

Service Parts Inventory (Maintenance, Repair, and Operations - MRO) Definition: Inventory used for supporting business operations but not directly used in production.

-

Example:

Aircraft spare parts stored by an airline for emergency maintenance.▪

Office supplies like printer ink and stationery.▪

-

Characteristics:

Used for repair, maintenance, and operational support.▪

Needed to minimise downtime in industries where breakdowns can be costly.

-

Inventory strategy: Holding a balance between cost and availability.▪

-

4.

Classification by Function (Purpose of Holding Inventory) This classification explains why businesses hold inventory.

Cycle Stock (Operational Inventory) Definition: The portion of inventory that is regularly used and replenished based on demand.

-

Example:

A bakery orders flour weekly because it is continuously used in production.▪

-

Characteristics:

Depletes over time due to customer purchases or production usage.▪

Regularly replenished through orders or production runs.▪

Influenced by Economic Order Quantity (EOQ).▪

-

1.

Safety Stock (Buffer Stock) Definition: Extra inventory held to protect against demand or supply fluctuations.○

Example:

A supermarket stocks extra bottled water before a predicted storm.▪

A car manufacturer keeps extra microchips in case of supply chain delays.▪

-

Characteristics:

Acts as an insurance policy against demand surges or supplier delays.▪

Helps prevent stockouts and lost sales.▪

Affects carrying costs, so should not be excessive.▪

-

2.

Pipeline Stock (In-Transit Inventory) Definition: Inventory that is currently moving through the supply chain but has not yet reached its destination.

-

Example:

Shipping containers full of clothes moving from China to Europe.▪

Aircraft parts being delivered to an assembly plant.▪

-

Characteristics:

Held due to transportation time and lead times.▪

Higher in industries with long supply chains (e.g., international trade).▪

Lead time reductions (e.g., faster shipping) can help reduce pipeline stock.▪

-

3.

Anticipation Stock (Seasonal or Hedge Inventory) Definition: Inventory stockpiled in advance to meet expected increases in demand.○

4.

Example:

Toy manufacturers increasing production before Christmas.▪

Clothing retailers stocking winter coats before winter.▪

-

Characteristics:

Helps businesses smooth out production instead of ramping up suddenly.▪

Used in industries with seasonal demand fluctuations.▪

Businesses may also buy extra stock before supplier price hikes.▪

-

Raw Materials

→ Work-in-Process

→ Finished Goods

→ Customers

Supplier Factory Warehouse Retail

Raw Materials Acquired from suppliers.

Work-in-Process (WIP) Moves through production stages.

Finished Goods Stored in warehouses or shipped to retailers.

Customer Demand Drives the flow of inventory.

Type Definition Example Key Benefit Potential Issue

Raw Materials (RM)

Unprocessed materials used in production

Iron ore for making steel

Ensures production continuity
 High storage costs
 Work-in-Process (WIP)
 Items partially completed in production
 Cakes being baked in an oven
 Reduces idle time in production
 High levels cause inefficiencies
 Finished Goods (FGI)
 Completed products ready for sale
 Cars at a dealership
 Available for immediate customer demand
 Risk of obsolescence
 Service Parts (MRO)
 Maintenance, repair, and operating supplies
 Spare aircraft parts
 Prevents operational disruptions
 Storage costs
 Cycle Stock Regular inventory for ongoing production or sales
 Supermarket groceries
 Predictable replenishment
 Dependent on supplier reliability
 Safety Stock
 Extra inventory to prevent shortages
 Extra microchips for car production
 Protects against uncertainty
 Increases carrying costs
 Pipeline Stock
 Goods in transit from suppliers to customers
 Imported electronics on a cargo ship
 Ensures supply chain flow
 Long lead times can delay supply
 Anticipation Stock
 Extra stock for expected demand increases
 Retailers stocking up for Black Friday
 Balances production and demand
 May lead to overstock if demand miscalculated
 Arguments for and against holding inventory
 Argument Explanation Example Implications
 FOR FOR FOR FOR
 Operational
 A minimum inventory
 A car assembly line
 Helps maintain a steady
 Necessity (Little's Law)
 level is required for smooth operations and to avoid disruptions.
 needs a steady supply of engines to prevent halts in production.
 production flow but requires careful management to avoid overstocking.
 Buffer Against Uncertainty
 Extra inventory protects against fluctuating demand or supplier delays.
 A hospital keeps an emergency supply of critical medications.

Prevents stockouts, but excess stock may increase carrying costs.

Exploiting Price Fluctuations

Bulk purchasing of raw materials when prices are low helps reduce costs.

A chocolate factory buys cocoa in bulk before expected price increases.

Can lower costs, but excessive buying can tie up capital.

Smoothing Production

Helps balance seasonal demand fluctuations without overburdening production capacity.

A toy company stockpiles inventory before Christmas.

Ensures product availability but requires forecasting accuracy.

Economies of Scale

Larger order quantities reduce per-unit costs and transportation costs.

A supermarket orders canned goods in bulk to get supplier discounts.

Reduces unit cost, but requires storage space and capital investment.

AGAINST AGAINST AGAINST AGAINST

High Capital and Holding Costs

Inventory ties up cash and requires warehousing, labour, and insurance.

A company spends millions storing excess electronic components.

Increases operational costs, reducing available cash for investment elsewhere.

Risk of Obsolescence & Deterioration

Perishable or fast-moving products can become unsellable over time.

A fashion retailer loses money when unsold clothing goes out of style.

Leads to waste and financial losses if demand is overestimated.

Hidden Operational Inefficiencies

High inventory can mask inefficiencies instead of fixing root problems.

A manufacturer stocks excess parts instead of fixing supplier issues.

Encourages poor process management and higher costs.

Longer Lead Times & Reduced Responsiveness

More inventory means longer processing and slower reaction to market changes.

A technology firm with old processors struggles to adapt to new trends.

Limits competitiveness and increases the risk of holding outdated stock.

Risk of Quality Defects

Defective products may remain undetected in large stockpiles.

A warehouse full of faulty car batteries leads to large recall costs.

Causes waste, potential reputation damage, and customer dissatisfaction.

Holding inventory has advantages and disadvantages.

Too much inventory leads to high costs, obsolescence, and inefficiencies. • Too little inventory can result in stockouts, lost sales, and production halts. •

Businesses must balance inventory to maintain smooth operations while minimising costs.

Hidden costs of inventory

Hidden Cost Explanation Example

Longer Lead Times

Excess inventory can slow down operations because processing large stock levels takes more time.

A warehouse with too much stock causes delays in retrieving and shipping items.

Reduced Responsiveness

Holding too much inventory makes it harder to adapt to market changes and customer preferences.

A smartphone company struggles to sell outdated models because they stocked too many units.

Hiding Operational Problems

Large inventory masks inefficiencies instead of fixing root issues like supplier delays or production bottlenecks.

A company holds excess raw materials instead of improving supplier lead times.

Quality Issues Not Identified Quickly

Faulty or defective items may go unnoticed in large stockpiles, leading to wastage and customer complaints.

A car manufacturer discovers defects in a batch of engines only after months in storage.

No Incentive for Process Improvement

Having large inventory buffers reduces the urgency to optimise production, supply chains, and forecasting.

A clothing retailer overorders instead of improving demand predictions.

Reducing excess inventory can help identify and fix underlying operational issues. A lean approach to inventory management (e.g., Just-in-Time - JIT) can reduce these hidden costs.

A reduction in inventory can be fatal.

Little's Law is a fundamental equation in inventory and queueing theory that describes the relationship between inventory levels, throughput rate, and lead time. It helps businesses understand how inventory affects operational efficiency and flow.

Where:

I Inventory (units in the system) The total number of items held in production or stock.

r Throughput rate (units per time period)

The rate at which inventory is processed or completed.

T Lead time (time spent in the system)

The average time a unit spends within the system.

It holds for all systems in a steady state:

Applies to manufacturing, logistics, healthcare, retail, and service industries. • Example: In a hospital emergency room, the number of patients at any time (I) is equal to the arrival rate of patients (r) multiplied by their average time spent in the ER (T).

•

Reducing lead time (T) reduces inventory (I):

If a company decreases the time taken to process or produce items, less inventory is required.

•

Example: Amazon's fulfilment centres use automation to reduce order processing time, decreasing inventory needed for the same throughput.

•

Increasing throughput rate (r) increases inventory (I):

If production speeds up but lead time remains the same, more inventory is present in the system.

•

Example: A car factory increasing production from 100 to 200 cars per day will have more inventory in progress unless lead times are reduced.

-

Balance between inventory, lead time, and output is essential:

Companies must optimize inventory levels without overloading the system. • Example: A fast-food restaurant needs just enough pre-cooked burgers to meet demand without excessive waste.

-

Practical applications of Little's law

Production and supply chain

Managing Work-in-Process (WIP):

Reducing lead time (T) minimizes WIP inventory without affecting throughput. ○

Example: A clothing factory that reduces sewing time per garment can produce more with less inventory waiting for processing.

-

-

Just-in-Time (JIT) Systems:

JIT focuses on reducing T to lower I, ensuring inventory is only produced when needed.

-

Example: Toyota reduces WIP inventory by delivering parts only as they are needed in assembly.

-

-

Service industry (healthcare, retail, and banking)

Hospital Example:

If a hospital has 200 patients (I) in its system and admits/discharges 50 patients per day (r), then the average time spent per patient (T) is:

-

-

Little's law

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If the hospital reduces T to 3 days, then the required beds (I) drops to:

-

Faster patient processing reduces the required bed capacity. ○

Retail Store Example:

If a supermarket sells 1,000 cartons of milk per day and has 5,000 cartons in inventory, the average time milk stays in stock is:

-

Reducing stock to 3,000 cartons without affecting sales means:

-

Shorter lead times free up storage space and lower costs.○

-

Applying Little's law to a computer assembly plant The computer assembly process has three stages:

Assembly1. Testing2. Packing3.

The entire process takes 975 minutes per unit.

Daily demand 1,600 units

Current WIP Inventory 4,800 units

Applying Little's law

1.

Each unit spends 3 days in production before completion. Reducing Inventory2. If the company wants to reduce WIP to 2,400 units, then:

By cutting WIP in half, lead time is also cut in half. Little's Law helps diagnose inventory inefficiencies. Reducing WIP can lower costs and increase efficiency. Balancing throughput rate and inventory optimizes supply chain flow.

Inventory performance metrics

To measure and optimise inventory efficiency, businesses rely on two key performance metrics:

Days of inventory (DOI) The number of days a company's inventory will last at the current demand rate.○

Formula:

-

Example:

A retailer has 10,000 units in stock and sells 1,000 units per day.

▪

The retailer has enough inventory to continue selling for 10 days before restocking.

▪

-

Interpretation of DOI Values

DOI Value Meaning

High DOI (e.g., 50+ days) Excess inventory, potential waste, high

-

1.

Moderate DOI (e.g., 10–30 days)

Balanced inventory levels

Low DOI (e.g., <5 days) Risk of stockouts, supply chain delays

Stock turnover ratio (stock turns) Measures how often a business replaces its inventory in a given period.○

Formula:

•

Example:

A company has an annual COGS of £1,000,000 and an average inventory value of £250,000.

▪

The company replaces its inventory 4 times per year.▪

•

Industry Benchmarks for Stock Turns

Industry High Turnover (Efficient)

Low Turnover (Inefficient)

Retail (Supermarkets) 20+ turns per year <5 turns per year

Automotive Manufacturing

10+ turns per year <3 turns per year

Pharmaceuticals 8–12 turns per year <4 turns per year

•

2.

High stock turnover means efficient inventory management. Low turnover indicates excess stock, leading to waste and obsolescence. Measuring DOI and stock turns helps businesses optimize inventory levels.

Suppose stock turnover is 5 times per year. Thus, each item sits in the warehouse for about 1/5 of a year. 22-40% represent realistic stock holding cost, including:

Handling• Cost of quality• Obsolescence• Warehousing•

The cost of holding inventory is hence approximately:

Stock turns is often used as key measure for operational and cost efficiency.

ABC Classification is a method used to prioritise inventory items based on their impact on overall costs and value. It follows Pareto's 80/20 Principle, which states that 80% of results come from 20% of causes.

Class % of Total Inventory Items % of Total Inventory Value

A Items 20% 80%

B Items 30% 15%

C Items 50% 5%

ABC classification breakdown

A-Class items (high priority)

High-value items that contribute the most to total inventory value.• Example: Car engines in an automotive factory.• Management Strategy:

Strict inventory control.○

Frequent reordering with lower safety stock.○

Demand forecasting and Just-in-Time (JIT) replenishment.○

•

B-Class items (moderate priority)

Mid-value items with moderate demand. • Example: Car seats in an assembly plant. •

Management Strategy:

Regular reviews of stock levels. ○

Less strict control than A-items but still monitored. ○

•

C-Class items (low priority)

Low-value items that make up most of the stock but contribute little to inventory value. •

Example: Nuts, bolts, and screws. • Management Strategy:

Simple, automated replenishment. ○

Bulk purchasing and infrequent reordering. ○

Low administrative effort. ○

•

Visualising ABC classification

The Pareto Chart shows the relationship between inventory volume and value:

% of Total Inventory Items % of Total Inventory Value

ABC analysis

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B-items 30% → 15%

A-items 20% → 80%

A-class: Strictly controlled, small percentage, high value. B-class: Medium monitoring, moderate value. C-class: Simple tracking, low value but large volume.

A dealership stocks cars, engines, and small components. How should each category be managed?

Inventory Type Classification Management Approach

Cars A Track each unit individually, prioritise fast sales.

Engines & transmissions

B Monitor stock levels, reorder regularly.

Nuts, bolts, fuses C Bulk purchase, automate replenishment.

Fixed order quantity models

Economic Order Quantity (EOQ): Determines the optimal order size that minimizes total cost.

•

Re-Order Point (ROP): Inventory level at which a new order is triggered. • Fixed time period models

Fixed Period Ordering: Orders are placed at set time intervals, quantity varies. • Lot-for-Lot

(LfL) Ordering (Order-Up-To, OUT): Orders are placed exactly based on demand.

•

Period Order Quantity (POQ): Orders are sized to cover demand over a specific number of periods.

-

Variable order quantity and ordering interval

Least Unit Cost (LUC): Orders are sized to minimize the per-unit cost. • Least Total Cost

(LTC): Seeks to balance ordering and inventory costs over several periods.

-

Part-Period Balancing (PPB): Advanced version of LTC, with adjustments for cost optimization.

-

Material Requirements Planning (MRP)

Calculates time-phased requirements for materials based on production schedules. •

The inventory cycle (sawtooth model)

Illustrates inventory fluctuations over time.

Order Placed → Inventory replenishment begins.

Lead Time (LT) → Time taken for replenishment.

Reorder Point (ROP) → Inventory level that triggers a new order.

Safety Stock → Extra inventory held as a buffer.

Ordering

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Fixed order quantity models

The Fixed Order Quantity Model is an inventory control system where a fixed quantity (Q) is ordered whenever the inventory level reaches a predetermined Reorder Point (ROP). The key features of this model include:

The order size remains constant each time an order is placed. • The time between orders varies, depending on demand fluctuations. • This model is useful for high-value items as it maintains low average inventory levels. • A continuous review system is required to monitor stock levels. •

Components of the fixed order quantity model: Reorder Point (ROP)

The Reorder Point (ROP) determines when a new order should be placed. It ensures that stock is replenished before it runs out, considering the lead time required for replenishment.

-

Where:

d = Average daily demand L = Lead time in days SS = Safety stock, used as a buffer against variability

-

If demand or lead time is uncertain, safety stock (SS) is added to prevent stockouts.

-

1.

Safety Stock (SS) Safety stock is extra inventory kept to account for demand fluctuations or delays in replenishment.

-

The amount of safety stock required depends on:

Demand variability▪

Lead time uncertainty▪

Desired service level (probability of avoiding a stockout)▪

-

Where:

Z = Z-score, corresponding to the desired service level
 σ_d = Standard deviation of demand per day
 L = Lead time in days

-

2.

Variants of fixed order quantity models:

(s, Q) Model (Reorder Point, Fixed Order Quantity)

The system orders a fixed quantity Q when inventory reaches reorder point s .○

Ensures stock is replenished as needed.○

Can incorporate safety stock.○

Example:

A retailer maintains an ROP of 50 units with an order quantity of 200 units. Whenever the stock drops to 50 units, the retailer places an order for 200 more units.

-

-

-

(s, S) Model (Reorder Point, Order-Up-To-Level)

Instead of ordering a fixed quantity Q , the system replenishes up to a predetermined level S .

-

The order quantity varies, but inventory is always replenished to S .○

Used when demand fluctuates significantly.○

Example:

-

-

A restaurant maintains a maximum inventory level of 500 kg of flour. When stock drops to 200 kg, they order enough to restore inventory back to 500 kg.

▪

Fixed order quantity models are implemented using automated inventory tracking systems to monitor stock levels in real time.

Barcode scanners and RFID tags help track inventory movement. • Enterprise Resource Planning (ERP) systems automatically trigger reorders when stock levels hit the ROP.

•

Two-bin and three-bin reordering systems are simple, practical methods to implement fixed order quantity models in small-scale operations.

Two-bin system Inventory is stored in two bins. ○

When the first bin is empty, a reorder is placed. ○

The second bin acts as a reserve to cover usage until the order arrives. ○

Used for high-consumption, low-cost items. ○

Example:

A hospital keeps two bins of syringes. When the first bin runs out, a new order is placed while using the second bin as a buffer.

▪

•

1.

Three-bin system Similar to the two-bin system, but includes a third bin for safety stock. ○

The third bin helps identify demand spikes (if the second bin empties before the new stock arrives).

•

Used for critical inventory items. ○

Example:

A manufacturing plant uses three bins of spare parts. If the second bin empties before replenishment, it signals the need for higher safety stock.

▪

•

2.

Fixed time period models

The fixed time period model is an inventory management approach where orders are placed at fixed time intervals, regardless of the current stock level. The order quantity varies depending on demand fluctuations, but the time interval remains constant.

Orders are placed at regular intervals (T), rather than when stock reaches a certain level. •

Order quantities (Q) vary based on demand since the last order. • Useful for items that require periodic restocking, such as groceries, fuel, or raw materials in batch manufacturing.

•

Larger average inventory levels are needed to prevent stockouts between order intervals.

•

Comparison with fixed order quantity models:

Feature Fixed Order Quantity Model Fixed Time Period Model

Order Timing When inventory hits reorder point At regular time intervals

Order Quantity

Fixed quantity (EOQ) Varies each time

Monitoring Continuous review needed Periodic review

Use Cases High-value or fluctuating demand items

Scheduled purchasing or vendor agreements

Basic Calculation: (T, S) Model

T = Fixed time interval between orders. S = Target inventory level to which stock is replenished.

Order quantity Q is determined by:

Where:

S = Target inventory level (pre-determined maximum stock level). I = Inventory on hand at the time of ordering. Q = Quantity to be ordered (varies each time).

A supermarket orders dairy products every Monday (T = 7 days). If the target stock level is 500 litres of milk (S = 500 l) and current stock is 200 L (I = 200 l):

Unlike EOQ models, the order size varies because demand is uncertain over the fixed time period.

Types of fixed time period models: Basic (T, S) Model

Orders are placed every T periods, and inventory is replenished to a fixed level S.○

Used when demand varies but can be estimated over time.○

Example: Restaurant Inventory System

A fast-food chain orders burger buns every 2 weeks (T = 14 days). If the target inventory is 2,000 buns (S = 2,000) and the current stock is 800 buns:

▪

•

1.

(T, s, S) Model (Modified) Orders are only placed if inventory falls below a minimum threshold (s).○

If stock remains above s, no order is placed.○

Helps avoid unnecessary ordering when inventory levels are already sufficient.○

Example: Automotive Parts Supplier

A company that stocks car batteries places orders every 3 months (T = 90 days), but only if inventory drops below 100 units (s = 100).

▪

If stock = 50 units, order up to S = 300. If stock = 120 units, no order is placed.

•

2.

Lot-for-Lot (Order-Up-To, OUT) Orders exactly match demand for the period, with no excess stock.○

Used in Just-In-Time (JIT) systems and zero-inventory strategies.○

Example: On-Demand Clothing Manufacturer

A t-shirt company only orders fabric when customer orders arrive. If 120 t-shirts are sold, fabric for 120 t-shirts is ordered—no surplus stock.

-

-

3.

Period Order Quantity (POQ) Modified EOQ model, adjusted to cover demand for a whole number of periods.○

Instead of a fixed order size (EOQ), the system orders to cover a set number of periods.

-

Example: Chemical Supplier

-

3.

A company with an EOQ of 90 units per quarter decides to order every 2 months instead.

-

Instead of 90 units per quarter, they order every 2 months based on demand.▪

Balances holding cost vs. ordering cost.▪

Variable order quantity and ordering interval

Variable Order Quantity and Ordering Interval refers to inventory control methods where:

The order size (Q) and/or ordering frequency (T) changes based on demand.• Unlike Fixed Order Quantity or Fixed Time Period models, orders are not placed at set intervals or for fixed quantities.

-

The goal is to minimize the total cost by balancing ordering costs and holding costs dynamically.

-

Adapts to demand fluctuations → Orders change based on real-time needs. Balances ordering and holding costs → Avoids overstocking while keeping ordering costs reasonable. Useful for irregular demand patterns → Ideal for seasonal products or unpredictable sales.

There are several ways to implement this approach: Least Unit Cost (LUC)

Orders are sized to minimize the per-unit cost (ordering + holding cost per unit).○

Formula:

-

Stops ordering just before the per-unit cost starts increasing.○

Period Demand Order Size Total Cost Cost per Unit

1 100 100 \$300 \$3.00

2 150 250 \$650 \$2.60

3 200 450 \$1150 \$2.55

4 250 700 \$1850 \$2.64

The best order quantity is 450 units because the cost per unit is lowest at \$2.55.○

1.

Least Total Cost (LTC) Orders are placed based on demand forecasts to balance total ordering and holding costs.

•

The method selects the order size where ordering cost $\approx$ holding cost.○

Example:

Setup cost per order = \$500 Holding cost per unit per period = \$1 Demand over 5 periods:

100, 150, 200, 250, 300 Finding the best order size that minimizes total cost over multiple periods.

•

2.

Part-Period Balancing (PPB) Advanced version of LTC where it considers future demand patterns.○

Uses "look-ahead" or "look-back" adjustments to reduce total cost.○

Avoids ordering too early if holding costs will be high.○

Example:

A retailer needs to order items for 4 upcoming months, with demand: Month 1: 200 Month 2: 300 Month 3: 400 Month 4: 500

•

PPB Approach:

•

3.

Instead of ordering each month separately, PPB balances future demand and orders a batch that minimizes total costs.

▪

When to use variable order quantity and ordering interval

Irregular demand products → Fashion, electronics, seasonal goods. • High-cost items →

Avoiding excess inventory cost. • Demand forecasting-based industries → Retail, distribution. • Supplier constraints → If order size restrictions exist. •

Comparison with other methods

Method Order Size Ordering Frequency

Best Used For

EOQ (Fixed Q) Fixed Variable Predictable demand

Fixed Time (T, S) Variable Fixed Regular purchases

LUC/LTC/PPB (Variable) Variable Variable Fluctuating demand

A key decision in inventory management is how much to order in each batch.

The trade-off in batch sizing

Using a large order size (i.e., ordering infrequently):

We suffer a large inventory holding cost. • Using a small order size (i.e., ordering frequently):

We suffer a large fixed cost of ordering• When ordering there are:
Clerical / labour cost of processing an order• Any fixed costs imposed by supplier• Inspection and return of poor quality products• Transport costs• Handling costs• Labour cost of organising transportation•

The challenge is to find the optimal order quantity that minimizes total costs.

Economic Order Quantity (EOQ)

The Economic Order Quantity (EOQ) formula calculates the optimal order size that minimizes the sum of ordering and holding costs.

Where: D = Annual demand rate CO = Ordering cost per order CH = Annual holding cost per unit

Key Insights:

Higher demand increases EOQ. • Higher ordering cost increases EOQ. • Higher holding cost decreases EOQ. •

Total cost formula

The total cost per period consists of:

Batch size decisions

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Where: D = demand per period Q = batch (lot) size CO = (fixed) cost of placing one order

CH = cost of holding one item in store for one period (financing and physical storage)

Ordering cost depends on number of orders per period:

Holding cost depends on average stock:

This gives the total cost per period formula, as a function of the batch size ordered (Q):

At the optimal order quantity (EOQ), ordering and holding costs are equal. Shouldn't the formula for total cost per period be:

Where: CV = variable cost (cost per item) SS = safety stock

Yes, though here it makes no difference to the optimal Q , so we often ignore the terms DCV and $SS \cdot CH$

Basic trade-off model:

At optimal order:

Reorder Point (ROP) and EOQ

To integrate EOQ with reorder point calculations:

Where:

R = Reorder point d = Average demand per time period (constant) e.g., $d = D/\text{working days}$ in the year L = Number of time periods between placing order and delivery

SS = Safety stock = $Z\text{-score} \times \sqrt{L} \times \text{Standard Deviation of Demand}$ EOQ determines batch size (Q) ROP determines when to order

A pharmacy sells 10,000 painkillers annually. Ordering cost per batch: £50. Holding cost per unit per year: £2.

They should order 707 units whenever stock reaches the reorder point.

A retailer expects to sell about 200 units of a product p.a. The storage space taken up in his premises by one unit of this product is valued at £20 per year. Interest rates are expected to remain close to 10% per year and one unit is bought at £100.

If the cost associated with ordering is £35 per order, what is the economic order quantity?

•

For administrative convenience, we can only order in minimum order quantities of multiples of 10. What is the total cost in case of ordering 20, and 30?

•

How does the EOQ change if we assume a more realistic 20%, or even 40% inventory holding cost, in addition to storage?

•

Standard EOQ cost function:

Now we replace CV by the new unit cost:

Where:

Range of Q Value of C

To find Q^* : Determine an algebraic expression for $C(Q)$ and for $C(Q)/Q$ in each price interval. 1. For each price interval find the feasible value Q_j^* that minimizes $T(Q_j)$. 2. Select minimum $T(Q_j)$ over $j = 0, \dots, n$. Call this $T^*(Q_j)$. Then:

$Q^* = \text{value of } Q_j^* \text{ corresponding to } T^*(Q_j)$ 3.

EOQ Variations: Discounts and Production: EOQ with all-unit volume discounts

Larger order sizes reduce the per-unit price.○

A balance between purchase cost savings and increased holding costs is needed.○

1.

EOQ with Incremental Discounts Discounts apply only to additional units purchased above certain thresholds.○

Requires calculating the total cost at different order levels.○

2.

What if we are producing the batch ourselves, rather than ordering it in from an external supplier?

Most of the issues are the same as they were when we were ordering the batch in. The differences are:

The cost of ordering becomes the cost of setup Clerical / labour cost of setting up a machine▪

Loss of production while set-up takes place▪

Return of poor quality products after start-up▪

1.

The batch does not now arrive instantaneously2.

•

The optimum batch size is known as the Economic Production Quantity (EPQ) •

Economic Production Quantity (EPQ)

Assume a constant production rate of $P > D$ for each batch.

Used when items are produced in-house rather than purchased. Analysis is as for EOQ, except CH is replaced by $CH(1 - D/P)$ reflecting lower average stock held over the period.

The batch size must account for production rate.

Where: P = Production rate per period. D = Demand per period. CS = Setup cost per production run. CH = Holding cost per unit per period.

Maximum inventory:

Assume a constant production rate of $P > D$ for each batch:

Average inventory:

Key Difference from EOQ:

EOQ assumes instantaneous replenishment. • EPQ considers gradual stock buildup. •

Category EOQ (Economic Order Quantity) EPQ (Economic Production Quantity)

Benefits Minimizes total inventory costs by balancing ordering and holding costs.

Minimizes total production costs by balancing setup and holding costs.

Simple to calculate and implement with basic inputs: demand, ordering cost, and holding cost.

Reduces frequent production setups, avoiding machine downtime.

Reduces stockouts and overstocking by determining the optimal order size.

Ideal for in-house production, where inventory builds up gradually.

Robust to errors in its parameters, mitigating risk by being less sensitive to deviations.

More realistic than EOQ since it accounts for gradual stock accumulation.

Reduces administrative effort by setting a repeatable order policy.

Balances production and demand, preventing excess inventory buildup.

Effective for bulk purchasing when supplier discounts do not apply.

Suitable for industries with high production setup costs (e.g., manufacturing).

Problems Assumes constant demand, which is unrealistic in real-world scenarios.

Requires production rate tracking, increasing administrative burden.

Does not account for lead time variability, making stockouts possible.

Ignores supply chain fluctuations, which may affect production.

Does not consider bulk order discounts, leading to suboptimal purchasing decisions.

Not suitable for JIT (Just-in-Time) production, as it encourages large batches.

Assumes instant replenishment, which is impractical in many supply chains.

Requires stable production schedules, which may not always be possible.

Ignores supplier constraints, such as minimum order quantities (MOQ).

Inventory buildup during production may increase holding costs.

Not suitable for multi-product ordering or shared supplier shipments.

Fails to adapt quickly to sudden demand changes, leading to overproduction.

Best for Retail, wholesale, procurement, and companies with predictable demand patterns.

Manufacturing, assembly, production-based industries with batch processing.

Not ideal

for

JIT (Just-in-Time) and Lean systems, where inventory minimization is key.

Highly unpredictable demand, where production should be more flexible.

EOQ vs. EPQ: When to Use Which?

Scenario EOQ EPQ

Purchasing finished goods from suppliers Yes No

Manufacturing own products No Yes

Stable and predictable demand Yes Yes

Highly variable demand No No

Just-in-Time (JIT) production No No

Supplier offers bulk order discounts No No

High setup costs for production No Yes

Short product life cycles (e.g., fashion, technology) No No

A supply chain is a set of three or more companies directly linked by one or more of the upstream and downstream flows of products, services, finances and information from a source to a customer.

A supply network is a network of connected and interdependent organisations mutually and cooperatively working together to control, manage and improve the flow of materials and information from suppliers to end users.

Why are supply chains important?

Traditional thinking: competition is driven by the 4P's: Product, Place, Promotion, Price

•

A final product is not the sole achievement of the OEM• Customer experience is determined by supply chain: quality, cost, delivery• Significant proportion of value sourced from suppliers!• Supply chains are connected systems:
Competitiveness of one tier is a function of the supply and distribution functions of surrounding tiers.

-
-

“Value Chains compete, not individual companies!” (M Christopher)

Supply chain management (SCM)

SCM refers to processes and exchanges across MULTIPLE firms. Sourcing/purchasing: includes the processes associated with identifying material and Supply chains

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tracking to assess supplier performance.

Make or Buy?• SC Strategy• SC Collaboration•

Logistics: plans, implements, and manages for efficient, effective flow and storage of goods and services from the point of origin to the point of end consumption.

Where should each company be located?• How much should be distributed, where? •

Operations Management Supply Chain Management

Concern Main concern is to maximise the efficiency of the internal processes involved

Main concern is to manage inter-firm networks to maximise the value to the ultimate customer (“user/chooser”)

Focus Focal firm only Dynamic interplay between firms

Optimisation “Optimises” within-firm processes with respect to the impact on the focal firm, and immediate customer

Sub-optimisation of the operation at the local firm in order to support holistic strategy

"Design" decisions in supply networks:

How should the network be configured?• Make or Buy?• Where should each part of the network owned by the company be located?• How much to distribute, and to where?• What distribution strategy to use?•

Configuration “Choices”:

Number of tiers• Number of suppliers at each tier?• Number of alternative suppliers?• Level of control over tiers?•

Freedom in making these “choices” depends on many issues: company size, number of available suppliers, standard vs specialised components.

Number of Tiers: Dell Customer Side:

Disintermediation or “Cutting out the middle men”• Supply Side:

Modular product design:

Standardised slots and interfaces between different computer components, designed with suppliers

-

Reduce supply chain complexity and tiers○

-

Late customisation•

Number of Alternative Suppliers: Example: Nokia / Eriksson Philips Fire

Philips factory in New Mexico producing radio frequency chips• 40% of output went to Nokia & Eriksson• Lightning strike caused fire in 2000• Destroyed much of factory, contaminated much of stock• Limited alternatives globally available• Nokia acted faster to buy up all global stock• Losses => Eriksson went out of cell phone manufacturing business•

How should the network be configured?

Simplicity• Number of supplier tiers dependent on complexity of product• Number of customer tiers dependent on selling channels• Resilient vs Efficient Supply Chain considerations •

Network configuration

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Also known as the vertical integration decision. No single company does everything that is required to produce its products. Every company has its “core competency” (Prahalad & Hamel, 1990) This gives rise to outsourcing (Venkatesan, 1992), sometimes many indirect processes are also outsourced – business process outsourcing.

Every commercial boundary introduces an extra cost:

Deliveries• Inventory• Contractual• Asset specific investments• Monitoring •

which has to be weighed against the pros and cons of producing inhouse. ‘Make-Buy Decisions’ are complex. Purchasing costs are relationship-dependent.

Outsourcing

The obtaining of goods or services from an external supplier company Why is the outsourcing provider cheaper?

Economies of scale• Economies of scope• Labour costs• Licensing costs•

Buyer Seller

1. Expensive tendering process Reluctance to invest or become dependent
2. Expensive contracts and monitoring Contractual pedantry
3. Expensive contingency plans Buck passing

Pros Cons

Focus on core competences• Harness lower labour cost at supplier• Access to technology• Stable and predictable financial planning in fee-for-transaction services

•

Loss of control over process• Limited ability to improve processes• Risk of opportunistic behaviour of supplier• Loss of human capital and tacit knowledge•

Make or buy

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Less investment risk •

Vertical integration

Merging together of two or more businesses that are at different stages of production e.g. a food manufacturer and a chain of supermarkets. Different than horizontal integration (acquisition/merger with a competitor). As span of control increases so should power in the marketplace.

Lower uncertainty, better coordination, lower transaction costs• Higher internal coordination required, weaker motivation for improvement?•

Monopolization? Accountability?

Reliance Industries (India):

Annual revenue - >\$100 billion• Largest private sector company in India• Activities span: Exploration and Production of oil and gas○

Petroleum refining and marketing○

Petrochemicals (polyester, fibre intermediates, plastics and chemicals)○

Textiles○

Retail ○

-

Largest polyester yarn and fibre producer in the world!•

Offshoring

The practice of basing some of a company's processes overseas to take advantage of lower costs. Outsourcing and offshoring are not mutually exclusive or inclusive.

Pros Cons

Lower labour costs• Development of new markets

-

Offset requirements•

Distance to existing customers• Customer perception• Not all jobs can be relocated• Long logistics lead-times, transportation and inventory cost•

Strategic risks:

Political• Currency• Intellectual property•

-

Environmental – off shoring may not be particularly green if your main market is at home

-

Impact:

The choice of location can have significant impact on profitability

e.g., retailing: a few metres can make the difference between profit and loss○

-

Driver:

Location changing decisions are prompted by:

Changes in demand for goods and services○

Changes in supply of inputs to the operation○

-

Objective:

To achieve an appropriate balance between:

Spatially variable costs of the operation○

The service level that the operation is able to provide to its customers○

The revenue potential of the operation○

-

So how to choose where?

Location choice: Absolut Vodka Absolut Vodka sells two-thirds of its production in North America. Yet when increasing its capacity, it chose to retain all production in Sweden. In the premium vodka market, quality is essential and authenticity is also important. Producing all vodka in Sweden ensures high quality and maintains the product's brand image.

Choice of country/region

Financial/regulatory Financial:

Exchange tariffs• Taxes• Shipping costs•

Regulations:

Employment• Environmental• Safety/Construction •

People/market Labour force:

Costs• Education/Skill•

Local culture:

Ethics• Language•

Proximity to Markets

Location

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Political stability• Economic stability• Social stability •

Infrastructure Infrastructure:

Transportation• Within country:

Access to ports• Location relative to transportation network• Location relative to supplies•

Multi-attribute utility (MAU) analysis

Country or regional decision using MAU analysis:

Identify the most important criteria1. Assign each criterion a weighting factor, F , between 0 and 1 such that:

2.

Give each location a score, S , on each criterion3. Multiply the score by the weighting factor4.

Calculate:

5.

for each location Choose the location with the maximum score6.

Pros Cons

Covers all dimensions• Quantitative output•

Weight assignment?• Subjective• Opinion based•

Where to locate?

Centre of Gravity Method: Used to find a location that minimises transportation costs

Based on the idea that all possible locations have a “value” which is the sum of all transportation costs to and from that location

•

The best location, the one which minimises costs, is represented by the weighted centre of gravity of all points to and from which goods are transported

x_i The x coordinate of source or destination

•

y_i The y coordinate of source or destination

V_i The amount to be shipped to or from source or destination

Pros:

Objective○

Quantitative○

Useful for searching for sites○

-

Cons:

One dimensional○

May not be sites at CoG○

-

Load-Distance Method: Used to choose the best location among available choices.

Plot locations and destinations on a x-y grid map• Calculate load for each destination and distance from each potential location• Calculate weighted (load * distance) sum for each location• Choose the location with the lowest load*distance

-

Pros:

Objective○

Quantitative○

Useful for evaluating candidate for sites○

-

Cons:

One dimensional○

Need target sites○

-

Example:

CoG:

Centre of gravity is (238,444) Hence choose site 3

Site X Y

1 360 180

2 420 450

3 250 400

Load-distance:

LD Site 1:

Similarly:

LD site 2 = 99791

LD Site 3 = 77546 Hence choose Site 3

The CoG/LD methods may not be optimal, but is a good place to start. In practice, it will also depend on:

Transportation network (actual effective distance)• Local infrastructure and incentives•

Demographic factors• Supplier base / alternative suppliers• Preferences of senior

management• Market•

How much to distribute, where? Simulation Optimisation Heuristics Let's see a heuristic: "NorthWest Corner Method"

Simple "rule of thumb" that seems to work• Easy to understand (common sense)• Can be efficient• Not necessarily optimal (but "good enough")• May not be evident why they work• The basic principles of the North West corner approach for allocating supply to demand. Create a matrix of sources and destinations• Set initial allocation from NW corner• Calculate the change in cost of supplying one unit from each currently empty cell while preserving demand/supply rim conditions (opportunity cost)

•

Reallocate the maximum possible quantity (subject to rim conditions) to the lowest cost cell, following the path evaluated.

•

Solution is optimal if all the opportunity costs are zero or positive• Limitations of algorithm: Heuristic - feasible but not optimal• Sensitive to starting point selected• Complex for multiple item types• No transport variations• Fixed supply / demand•

How much do we distribute, from which factories to which distribution centres, given associated costs, supply and demand constraints?

NorthWest corner method

Create a matrix of sources and destinations:

Aberdeen (V)

Preston (W)

Sheffield (X)

Ipswich (Y)

Winchester (Z)

Supply

Glasgow

18 16 12 28 54 46

Distribution

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Leeds (B) 24 40 36 30 42 20

Swindon (C)

22 12 16 48 44 34

Demand 27 16 18 10 29 100

Non-bold is the cost, bold is the quantity. Select the NW corner as start point1. Compare supply in row and demand in column. Allocate units equal to supply or demand – whichever is lower

2.

If demand in column is satisfied, move to square in right column (supply > demand) If supply in row is finished, move to square in next row (demand > supply)a. If both are satisfied, move to diagonal square formed next by row and next column (demand = supply)

b.

3.

Repeat until supply is exhausted and demand is satisfied4.

Total cost:

Finding an improved solution

Calculate the change in cost of supplying one unit from each currently empty cell while preserving demand/supply conditions (= opportunity cost)

Determine a closed path, starting at the empty cell being evaluated, and stepping from squares with assignments back to the original cell.

a.

Right angle turns in this path are permitted only at cells with assignments and the original empty cell.

b.

Because only the cells at the turning points are considered to be on the closed path, both empty and assigned cells maybe skipped over.

c.

1.

Reallocate the maximum possible quantity to the lowest cost cell, following the path evaluated.

Add and subtract the transport costs of each cell on the closed path as necessary to preserve the supply/demand conditions

a.

i.e., If you add a unit in one cell, then you should subtract a unit from the next cell on the closed path, so that the row and column totals remain unchanged

b.

2.

Back to Step 1. Solution is optimal if all the opportunity costs are zero or positive³.

Step 1: Calculate the change in cost of supplying one unit from each currently empty cell To use Leeds' supply to fulfil Aberdeen's demand:

Where can the numbers change to fulfil the demand? Possible improvement:

To use Swindon's supply to fulfil Aberdeen's demand:

Possible improvement:

Step 2: Reallocate the maximum possible quantity to the lowest cost cell, following the path evaluated.

Total cost:

Step 3: Solution is optimal if all the opportunity costs are zero or positive

Final optimal distribution cost, as they are all plus:

Issues in distribution problems

Unequal supply and demand:

If supply > demand, create dummy distribution points with zero transportation cost to represent surplus supply

•

If demand > supply, create dummy factory with zero transportation costs to represent unsatisfied demand

•

Total demand 71

Total supply 100

Add an extra “demand” point with zero cost to absorb surplus capacity.

Degeneracy in distribution problems:

Happens when the number of filled cells is less than the number of rows plus the number of columns minus one ($m + n - 1$)

-

Prevents cell evaluation• Can be solved by regarding one of the unallocated squares as having an infinitely small ε allocation

Does not affect constraints○

Treated exactly the same as other allocations○

But makes it possible to meet $m + n - 1$ condition ○

-

How do you make your products available to your customers at minimum cost, damage and time?

Direct shipping

Advantages Disadvantages

Point to point delivery• Mostly used for perishable items, specialty products, large bulky items

- Trucks may not be full, not best for high volume goods

-

Might not be best option if factory is remote

-

No risk pooling•

Warehousing

Remote production, variable finished-goods demand

Sort, store, pick, load

Advantages Disadvantages

Storage until demand• Enables risk pooling• Companies can share same resources• Some goods may require this e.g. goods with maturation point• Some clients may require this e.g.

JIT• Allows companies to work with optimal transport and order quantities• Allows companies to get closer to customers with less cost•

Obsolescence risk

-

Handling errors• Inventory costs – may vary from warehouse to warehouse

-

Cross-docking

Mass retailers

Distribution strategy

Wednesday, 19 March 2025 11:09 AM

Concepts Requirements

Items distributed continuously from suppliers through warehouses to customers

-

Need dedicated space and a fleet of vehicles• Warehouse rarely keeps items for more than 10-15 hours – minimal storage costs

-

Less congestion/transport costs at retailer• Handling• No risk pooling•
Close linkages between all participants in supply chain – needs IT support

-

Fast transportation• Large distribution volumes to justify cost•
Transshipment

Use transport hubs to change means of transport e.g. fish to refrigerated vessel, containers onto a train, imported goods from a ship to a truck.

Advantages Disadvantages

Can be used to combine small shipments into a large shipment (consolidation), or divide a large shipment into smaller shipments (deconsolidation)

-

Both cross-dock and transshipment may include product customization such as labelling and reduces the cost of doing so in a warehouse

-

May enable risk pooling•

Includes further handling• Unlike cross-docking transshipment may require storage facilities
- i.e. no reduction in holding or transport costs

-

Is not as simultaneous as cross-docking is

-

Risk pooling: Serving excess demand at one location with excess inventory at another location

Yes No

Warehousing Direct shipment

Transshipment Cross-docking

How to calculate time-phased requirements?

$N = 1$

Project management○

-

N = low volume, high variety

Job shop○
Single machine (Moore's algorithm)○
Sequential machines (Johnson's rule)○
Parallel machines○

•

N = high volume / batch production, standard products with options
MRP (Push)○
JIT (Pull)○

•

N = very high volume, few or no variants
Control theory○

•

How to convert customer orders into:

A production schedule for the multiple process stages on the shopfloor (work orders)• A supplier schedule for the required materials and components (purchase orders)• A delivery schedule at the customer interface (delivery promise)•

Two basic approaches:

Push scheduling - MRP• Pull scheduling - JIT•

Push and pull

Wednesday, 5 March 2025 11:09 AM

MRP & Dispatching Rules (Traditional Push System)

Planning: Based on forecasts. Orders and production are scheduled in advance.• Priority

Management: Use dispatching rules to decide which job to process next.• Batching: Larger batches optimize setup times, aiming to reduce costs per unit.• Flexibility: Low - the system follows a plan.• Quality Management: Defects can hide within large inventories, delaying detection.• Strength: Works well in stable environments with predictable demand.•

Weakness: Poor response to sudden demand changes or process problems.•

Push scheduling

In a push system, the orders are planned and issued centrally. Upon completion, the order is moved forward, until the next process is issued with the order to start processing it. Hence, the longest lead time process sees the new order first.

If you have a number of products, when should we make each component of each product, and when should we reorder raw materials? EOQ calculation assumed constant rate of demand for products throughout the year. But:

Demand may be "seasonal"• Even if demand for product is constant,•

if different products have common components, the demand for the components may be different. We need a systematic approach to plan production & ordering.

MRP systems were invented in 1960s to cope with computational complexity of scheduling
Need to plan at top level-item, for all items made in the plant• What happens if components are used in several final products?•

Computerised inventory control & production planning system

'A set of techniques which uses the bill of materials (BOMs), inventory data and the master production schedule to calculate future requirements for materials. It essentially makes recommendations to release material to the production system.'

Material requirements planning (MRP)

Wednesday, 5 March 2025 11:13 AM

Master production schedule

Drives MRP process with a schedule of finished products:

Outlines production schedule for top-level items only• Quantities may consist of a combination of customer orders & demand forecasts• Quantities represent what needs to be produced, not what can be produced•

MPS Period

Item 1 2 3 4 5 6 7 8

Clipboard 86 93 119 100 100 100 100 100

Lapboard 0 50 0 50 0 50 0 50

Lapdesk 75 120 47 20 17 10 0 0

Pencil Case

125 125 125 125 125 125 125 125

Bill-of-Materials (BoM):

Explosion of requirements for each component and raw material item• BOM enables a structure for the MRP calculation to be put into place•

Inventory master file:

Description Inventory Policy

Item Board

Item no. 7341

Item type Manuf.

Value class B

Buyer/planner RSR

Vendor/drawing 07142

Lead time 2

On-hand stock 53

Holding cost 1

Ordering/setup cost 50

Safety stock 25

Reorder point 39

Minimum order qty 100

Maximum order qty 500

Annual demand 5,000

MRP logic

Determine net requirements by

Adding allocations Specifically reserved future orders

To gross requirements Total independent and dependent demand before netting on-hand inventory and scheduled receipts.

Minus scheduled receipts Production orders and purchase orders that have already been released.

Minus projected on-hand On-hand inventory, projected into the future.

Equals net requirements

MRP Example (LT = 1, MOQ = 50):

Week 1 Week 2 Week 3 Week 4 Week 5

Gross requirements 90 85

Allocations 0 25

Scheduled receipts 40

On hand 20

Net requirements

Planned order releases
 Week 1 Week 2 Week 3 Week 4 Week 5
 Gross requirements 90 85
 Allocations 0 25
 Scheduled receipts 40 50 100
 On hand 20 60 20 10 10
 Net requirements 30 90
 Planned order releases 50 100

The ABC Picture Frame Co. wants to make 30 picture frames per week with the following product structure (indented bill of materials):

X Picture Frame
 Y Subassembly (2 required)
 Z Fastener▪

•

Z Fastener○

•

The company currently has 30 Picture Frames, 120 Subassemblies, and 70 Fasteners on hand. There are two scheduled receipts: an order for 100 Subassemblies to arrive at the beginning of week 2, and an order of 70 Fasteners to arrive at the beginning of week 3. The lead time for Picture Frames is 1 week, with an EOQ of 45. The lead time for Subassemblies is 2 weeks, with an EOQ of 70. The lead time for Fasteners is 1 week, and are ordered on a lot-for-lot basis. Construct the MRP records for parts X, Y, and Z for the next six weeks.

Part X: LT = 1 EOQ = 45

X 0 1 2 3 4 5 6

Gross requirements 0 30 30 30 30 30 30

Scheduled receipts

On hand 30

Planned order releases

Goes to:

X 0 1 2 3 4 5 6

Gross requirements 0 30 30 30 30 30 30

Scheduled receipts 45 45 45 45

On hand 30 0 15 30 0 15 30

Planned order releases 45 45 0 45 45 0

Part Y: LT = 2 EOQ = 70

Using part X as a guide (2 Y for 1 X):

Y 0 1 2 3 4 5 6

Gross requirements 0 90 90 0 90 90 0

Scheduled receipts 100

On hand 120

Planned order releases

Therefore:

X 0 1 2 3 4 5 6

Gross requirements 0 90 30 30 30 30 30

Scheduled receipts 45 45 45 45

On hand 30 0 15 30 0 15 30

Planned order releases 45 45 0 45 45 0

Hence:

Y 0 1 2 3 4 5 6

Gross requirements 0 90 90 0 90 90 0

Scheduled receipts 100 70 70

On hand 120 30 40 40 20 0 0

Planned order releases 0 70 70 0 0 0

Part Z: LT = 1 Lot for lot

MRP output

MRP systems can be used to generate a wide range of outputs, as they describe the future operation of the factory. These include:

Inventory order action: details of when orders for raw materials must be placed• Factory schedule: broad timing information on what should be occurring throughout the factory over the planning time horizon

-

Machine loading information: data on the quantity of work to be undertaken by each machine

-

Rescheduling data: given some discrepancy between reality and the prescribed schedule, the MRP system can indicate precisely what will be affected by late completion of some component

-

MRP is by far the most widely used production planning / scheduling system available today, but:

Assumes infinite capacity• Works with fixed lead times and fixed batch sizes• It has no feedback - having issued the plan, it assumes that this will work. This can be countered by re-calculating the schedule often to account for the current position

-

The accuracy of the data provided, including sales forecasting data, cannot be guaranteed. Keeping accurate records of inventory - both stock and in-process is notoriously difficult

-

Any delay in any component prior to an assembly operation will prevent completion of assembly - so shortages always deny the master production schedule.

-

MRP pushes production – production is not triggered by specific orders• MRP Systems inherently distort demand patterns in the supply chain•

MRP alternatives

MRP II is a method for the effective planning of all resources of a manufacturing company Manufacturing, marketing, finance, and engineering• Key distinctions from MRP: Integrated system• Used by all functional areas to “run the business”• Efficient data management• Financial planning/cost information• Sophisticated “what if” simulation capabilities •

Enterprise Resource Planning (ERP) takes a whole-enterprise view:

Manufacturing• Marketing• Sales• Service• Design• Inventory• Procurement• Distribution• Facilities management• Quality• HR• Finance•

Aimed at complete integration of enterprise level information. Considers linkage to the “extended enterprise”:

Suppliers• Customers •

JIT – Just-In-Time (Pure Pull System)

Planning: Directly responds to actual customer orders - no forecasting. • Priority

Management: No need for complex rules - just produce what’s ordered. • Batching: Ideal is single-piece flow, minimizing inventory. • Flexibility: High - system adapts directly to demand. • Quality Management: Defects trigger immediate investigation and resolution. •

Strength: Lean, flexible, customer-focused. • Weakness: Requires highly reliable processes and suppliers to work smoothly. •

Production orders are based on replenishment• The demand signal is conveyed via kanban cards• The system is tightly controlled, as it is “fragile”, i.e. unable to cope with large swings in volume or product mix

•

Need to keep schedule variability within 5-10%•

Pull scheduling

Batch sizes are as small as possible (ideal would be single-piece flow. Relies on the use of physical inventory control cues (or kanban) to signal the need to move or produce new raw materials or components from the previous process. Often requires suppliers to deliver components using JIT. The company signals its suppliers, using computers or delivery of empty containers, to supply more of a particular component when they are needed.

Supply chain infrastructure needs to be set up to do this. Delivery windows may be measured in minutes.

•

Frequent, small deliveries are not particularly “green”• Results in significant reduction in waste associated with unnecessary inventory, WIP, and overproduction

Might not necessarily be true for suppliers•

Just-in-time (JIT)

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MRP Approach JIT Approach

Focus High capacity utilisation Focus on producing what is needed

Utilisation More at each stage Lower capacity utilisation

Production Extra goes into inventory No surplus in inventory

Inventory High which reduces the problems being solved

Low so problems are exposed

Stoppages More because of problems Fewer

Aspect Push Strategy (MRP) Push-Pull Hybrid Pull Strategy (JIT)

Trigger Forecasted demand Parts replenished by forecasts, final product made on demand

Actual customer orders

Inventory High inventory levels Inventory for parts only, not for finished products

Low inventory levels

Production Timing

Produced in advance Components pre-stocked, final assembly starts after order

Produced when needed

Lead Time Short for customers (product ready)

Moderate – parts ready, but final product assembled on order

Longer, since production starts with order

Flexibility Low – based on forecasts

Moderate – customization possible at assembly stage

High – reacts to real demand

Variety / Volume

Low variety, high volume

Moderate variety, moderate volume

High variety, low volume

Part and Routing Complexity

Relatively simple – standardized parts and routings

Moderate complexity – common parts with customized final assembly

More complex – customized parts and routings

Typical Use Mass production Assemble-To-Order (ATO), configurable products

Custom, made-to-order products

MRP vs JIT

Wednesday, 5 March 2025 12:21 PM

OPT – Optimized Production Technology (Constraint-Focused)

Planning: Based on actual and forecasted demand, but focuses on ensuring bottlenecks are always active.

•

Priority Management: Everything revolves around keeping bottlenecks busy. • Batching: Batches sized to optimize bottleneck performance. • Flexibility: Moderate — non-bottlenecks can have flexibility, but bottlenecks must stay loaded.

•

Quality Management: Quality issues at the bottleneck are serious, but defects elsewhere are tolerated if they don't block flow.

•

Strength: Excellent for complex processes with limited bottleneck resources. • Weakness: Requires careful bottleneck identification and management. •

OPT (also known as TOC: Theory of Constraints) is a further production planning/control/scheduling approach:

The main concept is the importance of bottlenecks • Idea:

Bottlenecks control the performance of a system: “An hour lost at a bottleneck is an hour lost to the system” (Goldratt)

•

Focus improvement on the bottlenecks, allow the bottleneck to control the activity of the system. A busy operation is not necessarily a productive one.

How to schedule operations to maximise throughput?

Protect throughput at bottleneck with buffer• Bottleneck becomes pacemaker for entire process• Every minute that is not used at the constraint is a minute lost to the whole process•

Lean/JIT TOC

Kanban based pull• Reduce variability• Inventory is waste• Any waste improvement is a good thing

•

Drum buffer rope based pull• Variability can't be eliminated protect against it with buffers•

There is a role for inventory to protect bottleneck• Only bottleneck improvements bring gains

•

Company produces products F, G and H using resources A, B, C and D, each of which have 50 hours a week of availability. Demand, sales price, and raw materials cost are given as below. How should the company develop a product mix using the TOC approach?

Products

Processing times F G H

Optimised production technology (OPT)

Wednesday, 5 March 2025 12:40 PM

Resources A 5 10 10

B 15 10 5

C 15 25 20

D 10 15 15

Demand 90 80 70

. F G H

Sales Price 60 75 90

Raw material cost 15 25 20

Productio

n Managers Options

MRP & Dispatching Rules JIT (Just-in-Time) OPT (Optimized

Production Technology)

When to

order?

Early - according to forecast demand. Orders are placed well in advance based on anticipated future needs.

According to actual demand. Orders are only placed when customer demand

Use actual/forecast demand to keep the bottleneck busy. Orders are planned around the needs of key

arises. constraints.

What to

do next?

Dispatching rules determine the priority of jobs at each workstation (e.g., first-come-first-served, earliest due date first).

No choice - just track demand and respond immediately to what customers want.

Keep the bottleneck busy. Non-bottleneck processes are subordinate to the bottleneck's needs.

Batch
size?

Economic Batch Quantity (EBQ) or similar approach used — batch sizes are calculated based on setup times and economic factors.

One-piece flow – each product is made in response to individual demand (ideal case).

Batch sizes are determined to keep the bottleneck running efficiently (EBQ for the bottleneck).

When to
start?

According to standard processing times combined with forecast demand. Start work based on anticipated lead times.

Work starts when demand occurs — no production without a customer order.

Start dates are linked to the bottleneck's schedule, ensuring upstream and downstream processes align with bottleneck needs.

Predicted
due date?

Based on standard times and scheduling calculations. Due dates rely on estimated lead times.

Immediate – the system assumes demand-driven work is produced in real time.

Due dates are determined according to the bottleneck's ability to process work.

Defects? Hidden by inventory — defects may be carried forward because excess inventory buffers hide problems.

Defects trigger immediate stoppage and resolution. Quality is built into the process.

Defects are only a problem if they affect the bottleneck. Other areas may tolerate minor defects.

SPT Rule Minimises average completion/flow time on one machine

EDD Rule Minimises maximum lateness on one machine

Moore's Algorithm Minimises the number of late jobs on one machine

MDD Rule Minimises average tardiness on one machine (heuristic)

Johnson's Rule Minimises makespan in a two machine shop

Flow time Time that a job spends in the system

Lateness

Tardiness

Makespan Time between first job starting and last job finishing

How do we calculate time-phased requirements? How do you commit resources between a variety of tasks that need to be completed?

Project management

Low volume, high variety -- Job shop

Single machine (Moore's algorithm) Sequential machines (Johnson's rule) Parallel machines

High volume / batch production, standard products with options

MRP (Push) JIT (Pull)

Very high volume, few or no variants

Control theory

Associated scheduling problem: assembly line balancing

In a job shop situation, different jobs have different routings and different processing times.

Complexity can be boiled down to simple cases such as just one machine, two sequential machines, or two parallel machines.

The objectives we might face in scheduling are:

Minimise average completion/flow time of a set of jobs• Minimise the maximum lateness•

Minimise the number of late jobs• Minimise average lateness/tardiness• Minimise

“makespan” (time between first job starting and last job finishing)•

For each objective, an algorithm or heuristic exists to help us achieve our objective

Algorithm Optimal solution

Heuristic Reasonable, but not necessarily optimal solution

Scheduling

Wednesday, 29 January 2025 11:13 AM

Minimising average completion time

Job A B C D

Processing Times 11 3 4 2

If we do jobs in alphabetical order, and start at $t=0$, the schedule is:

Useful notation:

The overall completion time (20) does not depend on the schedule, but we might want to minimise average completion time per job. For our schedule:

We can do better by putting jobs with Shortest Processing Times earlier:

For new schedule:

SPT Rule Minimises average completion/flow time on one machine

Flow time Time that a job spends in the system

Minimising maximum lateness

Job A B C D

Processing Time 3 3 4 5

Due Date 9 8 16 9

If we do jobs in alphabetical order, the schedule is:

For this schedule:

For this problem, it is impossible to avoid some job being late We might want to minimise the maximum lateness: to do this, put the jobs in order of Earliest Due Date:

For new schedule:

EDD Rule Minimises maximum lateness on one machine

Minimising the number of late jobs

We might want to minimise the number of late jobs (eg to minimise the cost of subcontracting them) Moore's Algorithm:

Schedule jobs by EDD1. If no job is late, go to step 62. Find the first late job (call it the k th job)3. From amongst jobs 1 to k , remove the job with the longest processing time4. Return to step 1 with one fewer job to consider5. The schedule is the EDD schedule - plus removed jobs (in any order)6.

Job A B C D E F G H I

Processing Time

2 7 3 8 4 4 6 8 5

Due Date 5 10 15 22 23 24 25 30 33

Jobs already listed in EDD order, so we start with this schedule

Job A B C D E F G H I

Processing Time

2 7 3 8 4 4 6 8 5

Completion Time

2 9 12 20 24

Due Date 5 10 15 22 23 24 25 30 33

Job E is the first late job. Of the jobs up to E, the longest is D, so remove job D and start again.

New EDD order is:

Job A B C E F G H I

Processing Time 2 7 3 4 4 6 8 5

Completion Time 2 9 12 16 20 26

Due Date 5 10 15 23 24 25 30 33

Job G is the first late job. Of the jobs up to G, the longest is B, so remove job B and start again

New EDD order is:

Job A C E F G H I

Processing Time 2 3 4 4 6 8 5

Completion Time 2 5 9 13 19 27 32

Due Date 5 15 23 24 25 30 33

No (more) jobs are now late. Put removed jobs on the end of the schedule (in any order)

Final schedule is:

Job A C E F G H I D B

Processing Time

2 3 4 4 6 8 5 8 7

Completion Time

2 5 9 13 19 27 32 40 47

Due Date 5 15 23 24 25 30 33 22 10

Moore's Algorithm: minimises the number of late jobs on one machine

Minimising average tardiness

Hence it could be negative there is often little/no benefit from finishing a job early. Instead, define:

We write:

Minimising average tardiness is a meaningful objective. How do we achieve it? We could try EDD; if there is only one late job, EDD minimises the average tardiness. But for more than one late job, there is no algorithm for this problem. However, there is a heuristic:

Modified Due Date (MDD) Rule Put jobs in order of increasing MDD

At time t , $MDD = \text{the maximum of the due date and the earliest time at which we can complete a job}$:

Where:

At $t = 0$, Modified Due Date = Due Date for each job j :

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

5 10 15 22 23 24 25 30 33

Here, A has the smallest MDD (5), and completes at $t = 2$

$T = 2$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X 10 15 22 23 24 25 30 33

Here, B has the smallest MDD (10), and completes at $t = 9$

$T = 9$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X 15 22 23 24 25 30 33

Here, C has the smallest MDD (15), and completes at $t = 12$
 $T = 12$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X 22 23 24 25 30 33

Here, D has the smallest MDD (22), and completes at $t = 20$

So far, all jobs have had $MDD = DD$, but time is moving on.

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X X 24 24 26 30 33

Here, E has the smallest MDD (24), and completes at $t = 24$

We could have chosen F instead

$T = 24$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X X X 28 30 32 33

Here, F has the smallest MDD (24), and completes at $t = 28$

$T = 28$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X X X X 34 36 33

Here, I has the smallest MDD (33), and completes at $t = 33$

$T = 33$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X X X X 39 41 X

Here, I has the smallest MDD (39), and completes at $t = 39$

When will all parts be finished?

$T = 39$

Job A B C D E F G H I

Processing Times

2 7 3 8 4 4 6 8 5

Due Dates 5 10 15 22 23 24 25 30 33

Modified Due Dates

X X X X X X X 47 X

Final job is H, which completes at $t = 47$. How to measure the quality of my schedule?

Final Schedule:

Original Due Dates:

Total tardiness is 36, the average tardiness is:

Makespan

For a job shop scheduling with more than one machine:

Makespan time between first job starting and last job finishing

Every job visits the same machines in the same order Problem: Minimise makespan in a 2-machine flowshop

Jobs A B C D E

Processing Time – Machine 1 6 7 4 6 2

Processing Time – Machine 2 5 6 6 3 4

For ABCDE:

Makespan = 32

The makespan on m/c 1 is fixed. Thus, to minimise overall makespan, we need to minimise makespan on m/c 2. Two ways to do that:

Minimise amount of time m/c 2 continues to process beyond m/c 1 1. Minimise amount of time m/c 2 needs to wait before starting to process jobs 2.

Last jobs in schedule should have short processing times on m/c 2 First jobs in the schedule should have short processing times on m/c 1

Johnson's Rule: From the jobs not yet assigned, find the job with shortest processing time on either m/c.1. Which machine does that processing time occur on?

If that shortest processing time occurs on m/c 1, assign the job to the next free slot in the schedule

a.

If that shortest processing time occurs on m/c 2, assign the job to the last free slot in the schedule.

b.

2.

Go back to Step 1 until all the jobs are assigned.3.

Job Shortest time Machine Position

E 2 1 start

D 3 2 end

C 4 1 start

A 5 2 end

B Between jobs already allocated

Makespan = 28

Johnson's Rule minimises makespan in a two machine flowshop

An assembly line consists of a sequence of operations such as fitting of components, painting, or inspection. Some operations modify a single part or item. Other operations assemble two or more parts together. Could be moving assembly lines, in assembly cells, or stationary in groups. Some assembly lines are coupled with automated or at least semi-automated stations. Lines generally run at a fixed speed:

every "X seconds, a product is finished"

Crucial question: how many people do we need to do the work? Naïve answer: n operations need n people

Some operations may take less time than others, so some operators will be idle part of their time. One person can cover more than one task.

Ideally, the time required for stations to carry out their operations should not vary much from one station to another, otherwise:

some stations will be partly idle• and/or some stations will be overloaded•

We say we need to “balance” the assembly line.

Precedence constraints

Some operations cannot be done before others. This is relevant for lines, but not necessarily cells. Arrow means that one operation has to be completed before another one can start:

Calculate production rate T (called the ‘cycle time’, e.g. one item every T minutes). T is the maximum time the item can spend per station.

Suppose we want to produce:

Minimum number of stations:

I.e. 4 stations

Assembly line balancing

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Balancing loss

Station 1 1 & 2 1 mins

Station 2 3 & 4 1.3 mins

Station 3 5 & 6 1.1 mins

Station 4 7 & 8 1.5 mins

Station 5 9 0.7 mins

Station 1 1, 2, & 3 1.3 mins

Station 2 4 & 6 1.5 mins

Station 3 5 & 7 1.4 mins

Station 4 8 & 9 1.4 mins

Heuristic methods

Try all possible allocations? In allocating operations to stations, choose ‘most deserving’ operation first, followed by second ‘most deserving’, etc.

Criteria, for example: Longest sequential chain of followers:

Operation Followers

1 4

2, 3, 4 3

5, 6 2

7 1

8, 9 0

Take the longest first, thus #4 (1.0) before #2 (0.4) before #3 (0.3):

Station 1 2 3 4

#1 (0.6) #4 (1.0) #5 (0.6) #8 (0.7)

#2 (0.4) #6 (0.5) #7 (0.8) #9 (0.7)

#3 (0.3)

Total 1.3 1.5 1.4 1.4

Loss = 6.7%

1.

Total number of followers:

Operation Followers

1 7

2, 3, 4 4

5, 6 3

7 2

8, 9 0

Take the longest first:

Station 1 2 3 4

#1 (0.6) #4 (1.0) #5 (0.6) #8 (0.7)

#2 (0.4) #6 (0.5) #7 (0.8) #9 (0.7)

#3 (0.3)

Total 1.3 1.5 1.4 1.4

Loss = 6.7%

2.

Where should specific pieces of equipment, people, and work centres be located? Importance of layout:

Requires substantial investments of money and effort• Involves long-term commitments• Has significant impact on cost and efficiency of short-term operations•

Three basic layout types: Fixed-position1. Process (or Functional)2. Product3.

Hybrid layouts combine features of these basic layouts

Fixed-position layout

A layout that requires the product (e.g. because of its weight, size or delicacy) to remain at one location. The equipment, machinery, plant and people who do the processing move as necessary. Examples include:

Civil engineering projects• Shipbuilding• Restaurants with table service•

Process (functional) layout

A layout that groups together machines, equipment or people with similar functions or goals.

Many products share usage of the same equipment. The order and length of processing at various steps will differ. Associated with:

Job shops• Batch processes• Service shops•

Product layout

A layout that dedicates equipment and workers to specific products on a linear route. Can be dedicated to a single product or can be mixed-model lines. When volumes for a particular product are high, resources can be specialized and designed for a specific product. Associated with:

Continuous flow• Service factories•

Process Layouts Product Layouts

Advantages

Resources are general purpose and less capital intensive

•

Faster processing rates•

Layouts

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Flexibility is high—can adapt easily to changes in product mix or volume

•

Capacity utilization is high because of the ability to perform multiple jobs/tasks on each piece of equipment

•

throughput)

Fewer changeovers, less material handling, more efficient

-

Disadvantages

Frequent changeovers result in lost production time

-

High inventory levels are necessary to compensate for variable output rates

-

Product throughput time can be very long • Production planning and control is challenging

-

Low flexibility • Low utilization for lower-volume products

-

High risk of layout redesign for products/services with uncertain life cycles

-

Hybrid layouts

Layouts that attempt to combine the advantages of process and product layouts by grouping disparate machines into work centres or “cells” to work on products that have similar shapes and processing equipment.

Cellular

manufacturing

Layout in which machines are grouped into a cell that can process items that have similar processing requirements. Each cell has a small product layout. Flexibility of process layout combined with efficiency of product layout.

Selecting a layout type

Understand capacity and space requirements Understand information flows Understand cost of people and product flows Think about introduction of new products/volumes/mix Select appropriate material handling equipment Consider safety, ergonomics, and regulations.

Objective Challenge/Issue

Quality Quality at customer -> Quality of materials, manufacturing, distribution, retail

Speed Service time (time customer waits) 1. Cycle time (time product moves through the SC) 2.

Dependability Ability to deliver goods on time and in full, regularly

Cost Balancing cost contributions

Resilience Response to disruptions and changes

Fisher 1997: Matching the objectives of a supply chain depends on the characteristics of the product to be produced

Product Features

Functional Predictable • Few changes • Low variety • Low margin • Long lead times •

Innovative Unpredictable • Many changes • High variety • High margin • Short lead times •

Lean versus agile

Lean supply chains

Best in high volume, low variety and predictable environments

Agile supply chains

Less predictable environments where the demand for variety is high – volume, mix, product

Characteristic Lean Agile

Focus Eliminate waste Customers and markets

Objectives

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Partnerships Long-term, stable Fluid

Key measure Output measure such as productivity and cost

Measure capabilities, focus on customer satisfaction

Profit margin Low High

Forecasting Algorithmic Consultative

Purchasing

policy

Buy Assign capacity

Logistics

planning

Stable, fixed period Instantaneous response

Lean ideas in a supply chain maximize profits through cost reduction while agility maximizes profit through providing exactly what the customer requires.

Hybrid strategy

Strategies Explanation Market conditions & operating environment

Pareto 80:20 Using lean methods for the volume lines, agile methods for the slow movers

High levels of variety; demand is non-proportionate across the range

Surge/base demand separation

Managing the forecastable element of demand using lean principles; using agile principles for the less predictable element

Where base level of demand can confidently be predicted from past experience and where local manufacturing, small batch capacity is available

De-coupling point

The aim is to be lean up to the decoupling point and agile beyond it

Possibility of modular production or intermediate inventory; delayed final configuration or distribution

Cross functional teams in La Coruna Design Department

Gather fashion trends:

Fashion shows○

University campuses○

EPOS data from store○

Cafes etc.○

•

Broad appeal goods come from far east (40%)• Fabric for the rest are procured from closer locations and held undyed/unprinted• Operations that enhance cost-efficiency through economies of scale are conducted in-house (such as dyeing, cutting, labelling and packaging)

•

All other manufacturing activities, including the labour-intensive finishing stages are completed by a specialised Spanish network of more than 300 small subcontractors

-

Subcontractors work exclusively for Zara's parent firm, Inditex and receive

-

technological, financial and logistical support

Stores replenished two times a week • 15 days design to products compared to industry average of 6-9 months •

Planning together

SellCo Ltd supplies BuyCo with 10,000 items a year. SellCo bears the cost of shipping the goods, which is £400 per delivery regardless of how many items are shipped at once.

BuyCo's ordering costs are £50 per delivery. Each item has a value of £1,000, and both firms use an annual inventory holding cost rate of 30%. Let's assume that SellCo produces these items in a steady, continuous flow, and BuyCo similarly consumes them at a constant rate.

From SellCo's perspective, Q , the optimal delivery quantity is:

From BuyCo's perspective, Q , the optimal delivery quantity is:

Total cost = inventory carry cost + order cost Total cost to SellCo with its optimal quantity:

Total cost to BuyCo with its optimal quantity:

If we take Q_{SellCo} :

If we take Q_{BuyCo} :

If we consider the total joint cost as a function of Q – what value of Q would produce the jointly optimal solution?

Differentiate w.r.t Q and set to 0 to find the optimum:

This gives a TC of:

Through collaboration, both companies can come to a joint agreement to improve total costs.

Both SellCo and BuyCo must bear higher costs than they would if they optimised

-

Collaboration

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locally

£51,031 vs £22,485 • Will they want to share their savings? •

Suppliers have clear favourites

Arms-length, price driven Vs Collaborative, trust-driven supplier relationships

Research shows that:

plants with fewer 1st tier suppliers have better quality but they are less robust • trusting relationships show better performance • belief in common destiny – improved resilience e.g. Aisin Fire 1997 •

Which carmaker would you like to work with?

Secrets of Japanese Success Long-term collaborative relationship

Trust and commitment, respect of the right of mutual existence ○

1.

Dual sourcing Component volume is adjusted according to performance ○

Constant positive pressure ○

2.

Continuous Improvement Collaboration with suppliers on operational improvement○
Annual cost reductions are realised in collaboration, not isolation○

3.

Operations and logistics Level production schedules to avoid spikes in the supply chain○

Delivery systems that can handle mixed-load, small-lot deliveries○

Disciplined system of JIT delivery windows at the plant; suppliers deliver only what is needed, even if this compromises load efficiency in transport

-

4.

Mechanisms for collaboration

The goal is closer, more informed and effective relationship between customer and supplier.

Approaches:

Supplier Relationship Management

Commodity Supplier:

Where little or no SRM (Supplier Relationship Management) activity is undertaken as the suppliers provide infrequent one-off goods or services.

-

A company like Office Depot supplying office stationery to businesses. These purchases are infrequent, low-cost, and require little relationship management.

-

-

Performance Supplier:

Where focus is placed upon cost and service levels, as the supplier is providing off-the-shelf goods or short- to mid-term services that are not strategically important and are provided from a competitive market

-

-

-

environment. FedEx or UPS as logistics suppliers for short- to mid-term delivery needs. Businesses use their services for convenience and cost-effectiveness without them being strategically integral to operations.

-

Development Supplier:

Where focus is placed upon continuous improvement to service levels and cost, as the arrangements are more mid- to long-term, with some strategic value.

-

Intel providing semiconductor chips to computer manufacturers. The relationship involves continuous improvements in product performance and cost efficiency, making it a mid-to-long-term strategic arrangement.

-

-

Partner:

Where strategic long-term suppliers are managed to secure supply and drive collaborative engagement with shared benefits.

-

Boeing and Rolls-Royce have a deep strategic partnership where Rolls-Royce provides jet engines for Boeing's aircraft. This relationship involves long-term collaboration, shared R&D, and a secure supply chain agreement.

-

-

Vendor Managed Inventory

Supplier decides on inventory levels for customer (within bounds) and manages inventory at customer site

-

This works because the supplier can then better schedule its own production/distribution

-

Financial responsibility is retained by supplier, reducing manufacturer/retailer's risk

-

Supplier might buy unused material○

Suppliers usually receive POS data○

-

Collaborative Planning Forecasting Replenishment (CPFR)

Combines the intelligence of multiple partners across the supply chain in the planning and fulfilment of customer demand

-

Core elements:

Information sharing▪

Coordinated production planning▪

Joint demand forecasting▪

Coordinated shipments▪

Coordinated promotions▪

-

Superdrug and Johnson & Johnson

Objective was to trim inventory▪

Order and sales forecasts compared each week by external company▪

13% average stock reduction▪

-

-

Collaboration:

Advantages Disadvantages

Greater efficiency• Reduced variability• Reduced contract costs• Mutual learning• Greater stability for supplier• Greater visibility of supplier for customer •

Cost of communication• Risk of opportunism• Reduced flexibility in supplier selection•

Information leakages •

Information Services BUSINESS LEVEL

Electronic Data Interchange ORDER LEVEL

Barcode/RFID item data PHYSICAL PRODUCT LEVEL

Information services

Access to procurement/forecast data. Creates market efficiency by increasing visibility.

Requires a close supplier/manufacturer relationship.

Electronic data interchange (EDI)

Electronic transmission of purchasing requirements using common proprietary formats

Including purchase orders, invoices, delivery notices, etc. • May use private communication networks • Input directly into trading partner's information systems • EDI comprises 85-90% of business to business transactions •

Item data

RFID/Bar Coding/QR What?

Transfer of item/part information between supplier and customer • How?

Standardised ID [GS1] • ID as a pointer to networked data bases •

Why effective?

Cheap • Ubiquitous • RFID non line-of-sight •

Where Used?

Barcodes everywhere • QR emerging • RFID in high value supply chains •

Communication

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Strategic (Long-term) Planning (Mid-term) Operational (Short-term)

Supplier Materials programme • Supplier selection •

Materials requirements planning

-

Contracts •

Personnel planning • Ordering / receiving materials

-

Manufacturer Plant location • Production system spec

-

Capacity planning •

Master production scheduling

-

Capacity planning•
Lot-sizing• Machine scheduling• Shop floor control•
Distributor Physical distribution structure

-

Warehouse location•
Distribution planning• Warehouse capacity planning

- Warehouse management

-

Transport planning•
Retailer Product programme• Strategic sales planning

- Mid-term sales planning• Short-term sales

planning

-

Customer Market Modelling• Category Sales Forecast• Sales Forecast•
Supply chain information systems
SRM – Supplier Relationship Management:

These applications focus on the relationship between the manufacturer and its suppliers (any two supplier-customer links in the chain) and the procurement process that takes place between them.

Streamlined procurement process○

Tracking parts specifications, prices, order processes etc.○

Supplier performance measurement and comparison ○

-

MRP/MRP II/APS – Advanced Planning and Scheduling:

These systems produce schedules for what to make, where and when to make, and how to make it while taking into account material availability, plant capacity, and other business objectives.

Inventory planning○

Production scheduling○

-

Manufacturing Execution Systems:

Concentrates only on executing production in a manufacturing facility. An MES generally produces short- term schedules and allocates resources.

-

Inventory Planning Systems:

These systems observe demand patterns, take inputs on forecasting, costs, margins, and service levels, and then produce a recommended stocking policy.

-

IT systems

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WMS – Warehouse Management Systems:

These execute inventory planning commands and run the day-to-day operations of a warehouse. They also keep track of inventories in a warehouse

-

Transportation Planning:

These systems perform the analysis to determine how, when, where, and in what quantity materials ought to be transported.

Comparisons of different carriers, 3PLs – Route planning ○

Freight planning○

Considers mileage, taxes, tariffs, etc.○

-

Transportation Execution:

These systems make transportation plans work. They are less analytical than the planning modules, but serve as an operational link to the planning tools

-

Demand planning and Revenue Management:

These systems help companies forecast their demand using historical data and any information regarding future demand.

-

Revenue management deals with using price discrimination to maximise revenue e.g. through promotion planning, price differentiation, etc.

-

CRM – Customer Relationship Management:

These applications automate relationship between sellers and buyers by providing product and price information.

-

They also enable customer and product information availability in real time so that salespeople can direct their efforts. Naturally, these systems often “talk” to SRM applications.

-

A supply chain is not an entity• Hence there is no supply chain IT system• Each IT system is owned by one company• Hence can only achieve

Data sharing○

Cooperation○

Supply chain optimisation○

-

When it suits one or more companies to do so regardless of the IT system• Little or no IT support for strategic SC decisions [data analytics coming]•